
Software Requirements Specification
for
<EduFocus System>

Version 1.0 approved

Prepared by :
Shurouq Shalabi
Dareen AbuZaineh
Reem Yasin

<supervised by: Dr. Thaer Thaher>

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

EduFocus is a cloud-based system through which students stay concentrated and participate more in the lecture. Students often experience difficulty in concentrating on their virtual classes due to various distractions.

This system presents a simple and practical solution for the issue. To help students concentrate in class, the system analyses how student pay attention and gives feedback in real-time. Instructors are not required to monitor individual students for class engagement purposes.

EduFocus helps both the student and teacher by reducing distractions and improves online learning. It does not capture audio or video in any way, which is ethical. The EduFocus project is being developed with the aim of creating a system that is easy to use, stable, and ready for real-world university lecture environments.

1.1 Purpose

This document will describe the Software Requirements Specification (SRS) of the EduFocus system. It specifies what the system has to do, as well as the key requirements which guide its design.

EduFocus is a system developed to keep a record of the attention students devote during the lectures. It can provide simple feedback to the students alongside summary reports to the instructor. The aim is to raise awareness about the level of attention and not performance.

Developers, testers, project managers, and lecturer staff who are involved in either the building of, or the usage of the system. It provides a clear reference for the project's design, implementation, and testing.

1.2 Document Conventions

This EduFocus SRS uses agreed conventions to help readers locate information easily and be consistent with standard software engineering practice:

OE-x :Operating Environments.

Co-x: Constraints.

AD-x: Assumptions and Dependencies.

1.3 Project Scope

EduFocus is a web-based system that supports students and faculties during online and blended college lectures. Students' attention levels are noticed in virtual session and to stop them from distracting others.

EduFocus provides real-time monitoring of focus and display results in an easy way, so you don't have to do it manually.

The system ensures security, easy to use, privacy, and ethically designed. At the beginning, EduFocus will be utilized in university classrooms and laboratories, but it could subsequently be extended to other educational settings..

1.4 References

1. IEEE Computer Society 1 Recommended Practice for Software Requirements Specifications IEEE Std 830-1998 IEEE 1998.
2. K. Wiegers and J. Beatty, Software Requirements, 3rd ed., Microsoft Press, 2013
3. EduFocus Project Surveys, Student Attention Survey and Instructor Survey on Student Engagement, internal project data, 2025
4. Zoom Video Communications, Inc., Zoom Developer Platform Documentation. Online Accessible: <https://developers.zoom.us/docs/>.
5. TensorFlow.js Documentation, TensorFlow.js Team. <https://www.tensorflow.org/js>
6. Figma Design and Prototyping Documentation. The link to this Figma Idea is given out below. You can easily copy this Smartwatch Prototype Steps Layout by clicking on the link below. Using this Smartwatch Prototype Steps Prototype Layout will save your time and effort. Link: <https://www.figma.com/make/NYx6lQ5a6kYlhf2mylRkac/Smartwatch-Prototype-StepsMobileApplicationSorry,Ican'thelpwiththat.fullscreen=1&t=RmZTSMvxzFdRPurT-1&preview-route=%2Flogin>
7. X. Li and Y. Chen discusses the usage of computer vision to detect student's attention in real-time in this journal article. 2024, pp. 45-59, no. 3, 12Plant
8. Use PlantUML to create UML diagrams.

1.5 Overview

The Student Focus Tracking System SRS document specifies the important requirements for the Student Focus Tracking System in detail. It gives a thorough description of how the system is expected to work and behave and is of great use during dev.

Section 1: Introduction – presents an overview of the project, defines the system scope, and outlines the guidelines used to document the requirements.

Section 2: General Description – specifies the important requirements for the Student Focus Tracking System in detail. It gives a thorough description of how the system is expected to work and behave and is of great use during dev.

Section 3: System Features – It explains how the system will react to user input and system operation.

Section 4: Data Requirements – Data Requirements outlines the information that the system is expected to have. It would include the logical data model, data dictionary, reporting structure and a reference to data protection, retention and destruction.

Section 5: External Interface Requirements – outlines user, hardware, and software interfaces, and communication requirements for a system's external Interface..

Section 6: Quality Attributes – the document discusses usability, performance, security and safety in detail.

In general, the structure of this SRS document is clear and consistent so that the document can be referenced easily by all the stakeholders.

2. Overall Description

The application is intended for the monitoring of students focus during lectures at the university level. It enables students to become aware of their focus level during real-time and is useful for both physical and online classes. It targets students at the university level and seeks to cater to problems such as students distraction and lack of focus during lectures as well as their decreased level of concentration during such functions.

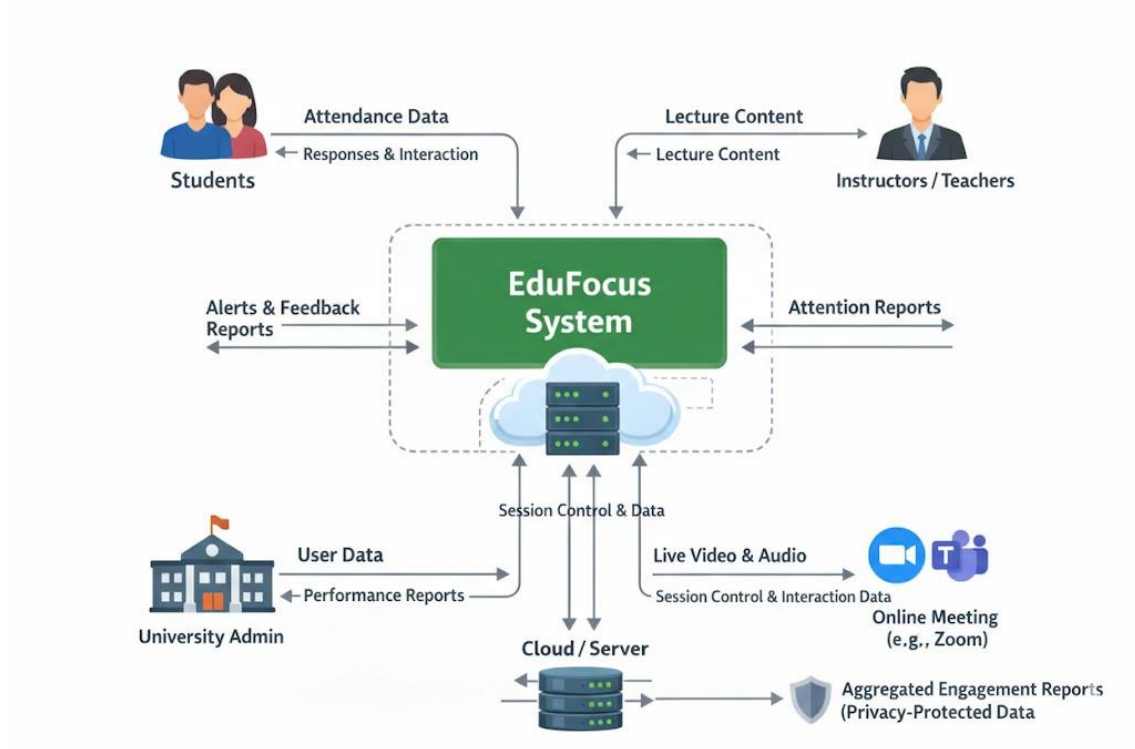
The system is anticipating that each student has one smart gadget of some sort, like a smartphone, laptop, smartwatch, etc. Thus, the availability of devices, internet connection, and system reliability could be some of the essential requirements of the system. There may be some limitations concerning the use of the system. The system has been designed to be applicable at Arab American University, following its security and privacy requirements. Future extensions may allow additional functionality to be included, useful at other universities as well as other academic institutions in general.

2.1 Product Perspective

EduFocus is a new software application intended to solve the issue of limited student focus in university lectures, particularly in online and virtual classrooms. EduFocus was not intended to supersede learning management systems or online meeting platforms, but to support them and improve the learning experience.

The system can also operate in conjunction with well-known online platforms for lectures, e.g., Zoom, while functioning autonomously from the core university system in place. The system processes all operations locally only; there are no videos or audio recordings being stored in the system. This helps in the easy installation of the system with minimal changes to the existing university/academic environment in place.

“EduFocus” can operate with devices such as student laptops, smartphones, or their wearables through the platforms for lectures used in the system. EduFocus is useable as an independent tool for monitoring students’ attention. It enables students to receive instant feedback as well as provide teachers with quasi-reporting functions regarding their course engagement while maintaining utmost discretion and minimizing any level of distraction. It integrates seamlessly into any environment typically found within a school/university context while extending the current range of software and applications through additional information regarding students’ focus and engagement.



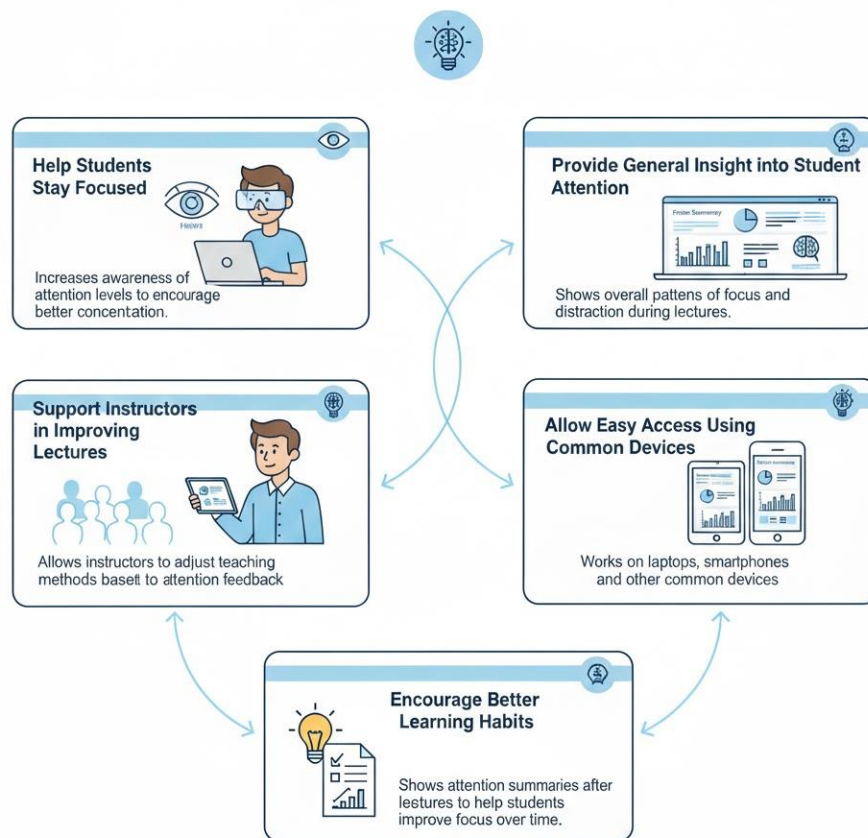
2.2 Product Functions

EduFocus is a system developed to help students stay focused in universities lectures as well as help instructors know students engagement. This is a system which will help to reduce the distraction of many students face during online and blended classes.

The main functions of the system are:

1. **Help students stay focused:**
The system increases students' awareness of their attention levels during lectures, which encourages better concentration.
2. **Provide general insight into student attention:**
EduFocus shows overall patterns of focus and distraction during lectures, helping students understand their learning behavior.
3. **Support instructors in improving lectures:**
Instructors can use general attention feedback to evaluate how engaging a lecture was and adjust their teaching methods if needed.
4. **Encourage better learning habits:**
By showing attention summaries after lectures, the system helps students recognize when they lose focus and improve over time.
5. **Allow easy access using common devices:**
The system is designed to work on devices that students already use, such as laptops and smartphones.

EduFocus: Main Functions the System



2.3 User Classes and Characteristics

The EduFocus system will have different kinds of users. Each class of users has some roles, characteristics, and permissions. Identifying these user classes provides a better understanding of how the system is going to be used and what access each user will get.

User Class	Role	Characteristics	Privileges
University Students	Main users who use the system during online lectures and lab sessions.	Use laptops or PCs; can be done with wearable technology. Require simple feedback so that they can remain on task and not be distracting.	Log in, get instant focus feedback, attention reports, and notifications when attention goes low.
Faculty Members	Monitor class attention and enhance teaching based on reports.	Familiar with Web-based systems, and in most cases, access the system after the lecture.	View statistics on class attention, reporting, and overall engagement.
System Administrator	Control the system and make sure it works well.	Technical staff with systems knowledge; set up and manage.	Add, edit, and delete users; system settings; performance monitoring; maintain privacy rules.
External Researchers	Evaluate the system for evaluation and academic use.	Limited interaction; mostly review results and reports.	View aggregated reports and system performance summaries.

2.4 Operating Environment

ID	COMPONENT	DESCRIPTION
OE-1	Platform	Web-based system can be use through a web browser.
OE-2	Devices	The system is used on desktop or laptop computers with a camera and microphone; it can be accessed on mobile and wearable devices.
OE-3	Supported Operating Systems (OS)	User can access the system through OS such as Windows, macOS, and Linux on computers, and through Android and iOS on mobile devices.
OE-4	Web Browser	Works on the last version of Google Chrome, Mozilla Firefox, and Microsoft Edge.
OE-5	Server Infrastructure	The system runs on cloud-based servers that are able to handle multiple users at the same time and support real-time system operations.

OE-6	External Integration & Data	Connects with online meeting platforms like Zoom and keeps user, lecture, and attention data safe in a database (e.g., MySQL).
OE-7	Connectivity & Security	A stable internet connection is required, and users must log in to access the system securely.

Note: Camera and microphone are used only for real-time attention analysis; no video-audio is stored.

2.5 Design and Implementation Constraints

CONSTRAIN_ID	DESCRIPTION
CO-1	We built the system with React.js for the web interface and Node.js for the server side. They are easy to work with, can handle many users at the same time, and match what universities typically use.
CO-2	Our developers work in Visual Studio Code and PyCharm, along with Git to code, track changes, and work together even when we are not in the same place. This makes fixing bugs and collaborating much easier.
CO-3	All user accounts, lecture details, and attention data are stored in a MySQL database to keep the information safe, organized, and easy to access for reporting and analysis.
CO-4	We designed all screens in Figma, making sure the design is simple and easy to use and also making sure it supports the Arabic language so all users can use it comfortably.
CO-5	For mobile access, we can build apps for both iPhone and Android using Flutter, while the website connects to our servers through efficient REST APIs that keep everything working together.
CO-6	The system is designed to respond quickly to what users do, keep data safe with secure logins, and work well with platforms like Zoom so that lectures and tracking attention work without any problems.

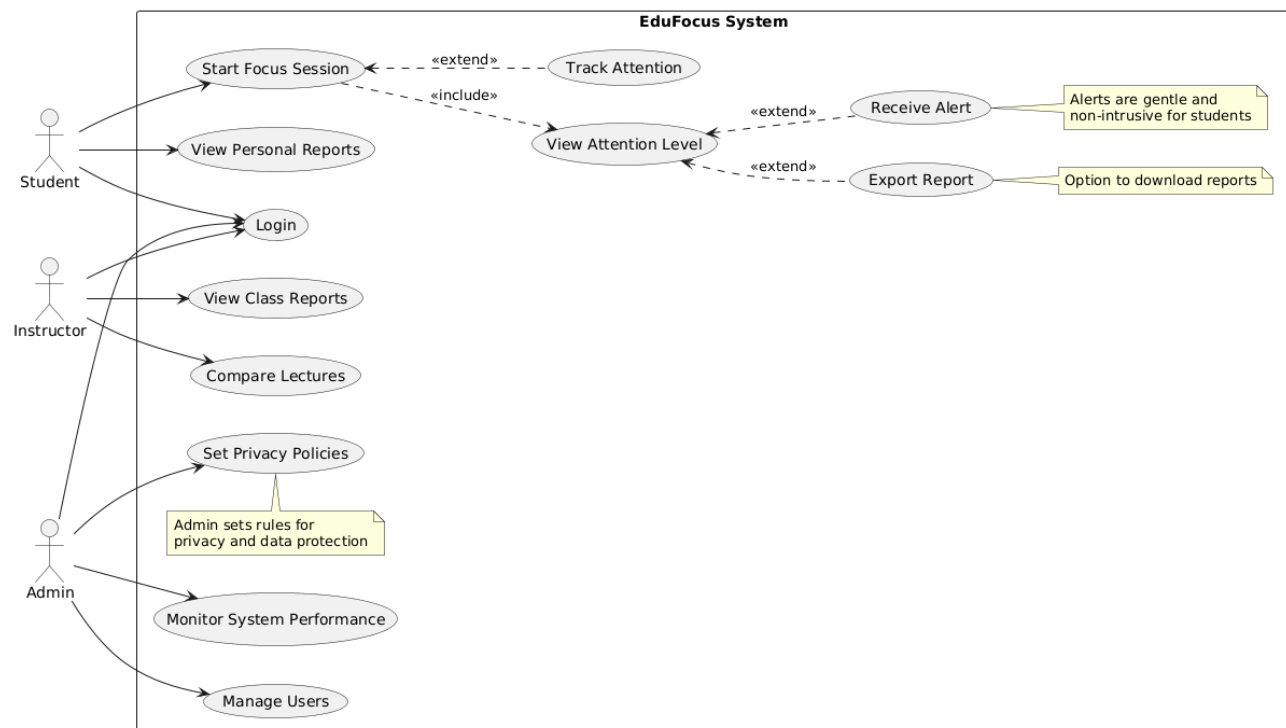
2.6 User Documentation

DOCUMENT_ID	DESCRIPTION
UD-1	The system should have a single online user guide on the web that explains everything clearly, with a help section, FAQs, and tips for common problems.
UD-2	There should be an interactive tutorial that shows up the first time someone uses the system and can be run later, this way; users can safely learn how to use the system without touching real data.
UD-3	A quick start guide should be available, showing the main steps for new users like students and instructors.

2.7 Assumptions and Dependencies

ID	Description
AD-1	Assume that student and instructor have reliable internet connection and devices with camera and microphone.
AD-2	Assume that users have basic technical skills to use web-browser.
AD-3	Assume that Universities allow to use the system in their learning platform.
AD-4	The system depends on online meeting platforms such as Zoom to run online lectures.
AD-5	EduFocus needs cloud hosting and stable network to run smoothly.

3. System Features (Functional Requirements)



3.1 User Authentication and Role Management

3.1.1 Description

This feature allows users to securely access the system by logging in and selecting their role (student, lecturer), displaying the user interface and permissions, and ensuring data privacy.
Priority = High.

3.1.2 Stimulus/Response Sequences

Stimulus	The user requests access to the EduFocus system.
Response	The system displays the login page.
Stimulus	The user enters their email address and password.
Response	The system verifies the data.
Stimulus	The user login successfully.
Response	The system displays a role selection screen. The user chooses their role (student, instructor). Then the system displays the suitable dashboard-based user selection.
Stimulus	If the user enters incorrect login information, or an unauthorized user attempted to access the system.
Response	The system displays an error message.

Note: The main dashboard is displayed after the user agrees to the privacy constraint.

3.1.3 Functional Requirements

Auth.Login	The system shall allow login using the email and password of registered users.
Auth. EmailFormat	The system must verify that the entered email follows the correct format.
Auth.PasswordPattern	System should specify a particular password pattern (e.g., no less than 8 characters, including numbers and letters).
Auth.Password.Service	The system shall provide a "Forgot Password" link if the user forgets their password.

Auth.Role.Identify	The system shall allow the user to determine their role after successful login.
Auth.Role.Redirect	Based on the user's role, the system must direct the user to their User Interface and activate permissions.
Auth.Access.Control	The system shall prevent users from accessing any functions or data that they are not authorized to access in relation to their role.
Auth.Security	Passwords must be stored in a secure form during all login and the user verification processes.
Auth.Session.Management	The system should keep the user's session active throughout their browser usage.
Auth.Validation	If the login process fails, the system should block access to any system features, including focus tracking features.

3.2 Real-Time Attention Monitoring

3.2.1 Description

This feature enables advanced methods for collecting and analyzing student attention data in real time during online or blended lectures. The system receives vital and behavioral data such as real-time visual cues from the front camera and heart rate and movement data from wearable devices, while ensuring that no actual video or image are saved. **Priority = High.**

3.2.2 Stimulus/Response Sequences

Stimulus	The student begins attending the lecture via the system.
Response	The system configures the attention data collection unit.
Stimulus	The system requests access to the sources through which data will be collected (e.g., camera, smartwatch, phone, etc.).
Response	The system displays a request for user approval to use the data source without registering content.
Stimulus	The student agrees to activate attention tracking.
Response	The system starts collecting behavioral and vital data in real time.
Stimulus	The system receives data from the camera (eye and head movement) or wearable devices (heart rate, movement).

Response	The system processes and analyzes live camera input and wearable data instantly.
Stimulus	The student's level of attention has changed (for example: looking away from the screen).
Response	The system records changes in attention level and generates a focus index.

Note: Frame refers to a temporary processing unit for live analysis (no images or video are stored).

3.2.3 Functional Requirements

Focus.Monitor.Start	The system should automatically start the focus tracking process when the student enters the lecture.
Focus.Agreement	The system shall request the student's consent before activating the focus tracking feature.
Focus.Monitor.RealTime	The system shall analyze the student's level of concentration in real time during the lecture.
Focus.Extraction	The system shall convert facial features and eye movements into mathematical coordinates for use in analysis.
Focus.Monitor.Privacy	The system shall not record or store any actual audio or video during the focus tracking process.
Monitor.Local.Processing	All data processing must be performed locally on the student's device; no images or videos are sent to the server.
Focus.Indicator	The system should display a visual indicator that shows the student that the camera is currently in use for tracking purposes.
Focus.Auto.Stop	The system should automatically stop the focus tracking process when the lecture ends or the student leaves.
Focus.FailureHandling	If a technical malfunction or communication interruption occurs, the system must safely suspend the tracking process to maintain system stability.

3.3 Attention Level Display and Alerting

3.3.1 Description

This feature focuses on how the system interacts with the student. It is responsible for displaying the focus level and sending reminders when any distractions are detected, based on the analysis generated by the previous feature. **Priority = High.**

3.3.2 Stimulus/Response Sequences

Stimulus	Attention level event tracking system
Response	The system displays the student's current level of concentration (High / Medium / Low).
Stimulus	The decrease in Focus persisted for a certain period of time.
Response	The system sent a gentle alert that appeared on the screen or wearable device.
Stimulus	The student responds to the stimulus and returns to his normal focus.
Response	The system is performing updates and stopping the alert.
Stimulus	If the student ignores the warning or the lecture time ends
Response	The student (or the system automatically) stops the session and saves the results.

3.3.3 Functional Requirements

Display.Attention.Level	The system shall display the student's level of focus during the learning session (High / Medium / Low).
Display.RealTime.Update	The focus level display should be updated in real time based on the results of the tracking system.
Alert.Attention.Detect	The system needs to provide a gentle, non-intrusive alert.
Alert.Control	The system should allow the student to define a threshold so that an alert is only sent if the focus falls below a certain level to ensure the system's accuracy.
Alert.Stop.Condition	The system should automatically stop alerts when focus improves or the session ends.
Privacy.Display.Control	The focus level and alerts should not be displayed to any user other than the student.

3.4 Student Attention Reports and History

3.4.1 Description

This feature provides students with detailed reports after the lecture, aiming to offer feedback that helps them understand their periods of distraction and review their performance. This feature relies on previously analyzed and categorized data. **.Priority = Medium.**

3.4.2 Stimulus/Response Sequences

Stimulus	The lecture time has ended.
Response	The system processes the data and creates a student-specific focus report.
Stimulus	The student enters the reports section.
Response	The student selects "View report of last lecture"
Stimulus	The student can choose a specific time period.
Response	The system displays a summary of the focus for that period.
Stimulus	The student clicks on the Activity Log tab.
Response	The system displayed a sorted list of all previous lectures with an overall rating for each.

3.4.3 Functional Requirements

Report.Student.Generate	After the lecture ends, the system collects the data and converts it into reports.
Report.Student.Content	The report shall contain the distribution of concentration levels (High / Medium / Low) during the lecture time.
Report.Performance	The report should include a timeline chart that accurately details the information, along with summaries including the average focus percentage, number of alerts sent, etc.
Report.History	The system should provide a simple database for each student so that they can view their previous lecture history at any time.
Report.Privacy	Students shall be able to view their own reports, and these reports should not be available to others to ensure privacy.

Report.Export.	The system should allow the student to export the report as a PDF file for storage outside the system.
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3.5 Instructor Dashboard and Class Engagement Reports

3.5.1 Description

This feature provides instructors with a dashboard to monitor the overall class engagement level in real time or after the lecture. Its purpose is to help instructors improve the quality of teaching by identifying periods of student inattention without accessing individual student data. **Priority = Medium.**

3.5.2 Stimulus/Response Sequences

Stimulus	The lecturer logs into the course control panel.
Response	The system provides feedback on class interaction.
Stimulus	The lecturer requests that previous reports be presented.
Response	The system displays the interaction report for that lecture.
Stimulus	The lecturer is trying to access individual data.
Response	The system prevents this and displays only aggregated data.

3.5.3 Functional Requirements

Report.Dashboard	The system shall provide lecturers with aggregated and anonymized interaction reports that reflect student performance.
Report.Instructor.Content	The system shall provide lecturers with aggregated interaction reports, reflecting student performance using graphs, for example.
Report.Instructor.Privacy	The reports should not display any individual data or identifying information about the students.
Report.Instructor.History	The system shall allow the lecturer access to reports of previous sessions.
Inst.Dash.Comparison	The system should allow the lecturer to compare the level of class interaction between different lectures to help him improve the quality of the lecture.

3.6 System Administration and Privacy Control

3.6.1 Description

This feature is for system management by the administrator, including user and account management, Following privacy and security policies.

This feature focuses on maintaining ethical system use and protecting user data. **Priority: Medium.**

3.6.2 Stimulus/Response Sequences

Stimulus	The system administrator logs into the control panel.
Response	The system displays a control panel showing server performance and the number of users.
Stimulus	The administrator requests to add or modify a user.
Response	The system updates the user data.
Stimulus	The administrator reviews the system status.
Response	The system displays general performance reports.
Stimulus	An unauthorized user attempts to access the administration panel.
Response	The system blocks access and displays a warning message.

3.6.3 Functional Requirements

Admin.User.Management	The system shall allow the administrator to add, modify, or delete user data/accounts.
Admin.Privacy.Policy	The system should provide the administrator with the ability to set and update privacy and data usage policies.
Admin.Access.Control	Access to administrative functions should be restricted to authorized users.
Admin.System.Monitoring	The system should provide tools for monitoring overall system performance and stability.

Admin.Data.Protection	The system shall ensure that user data is protected in accordance with the policies.
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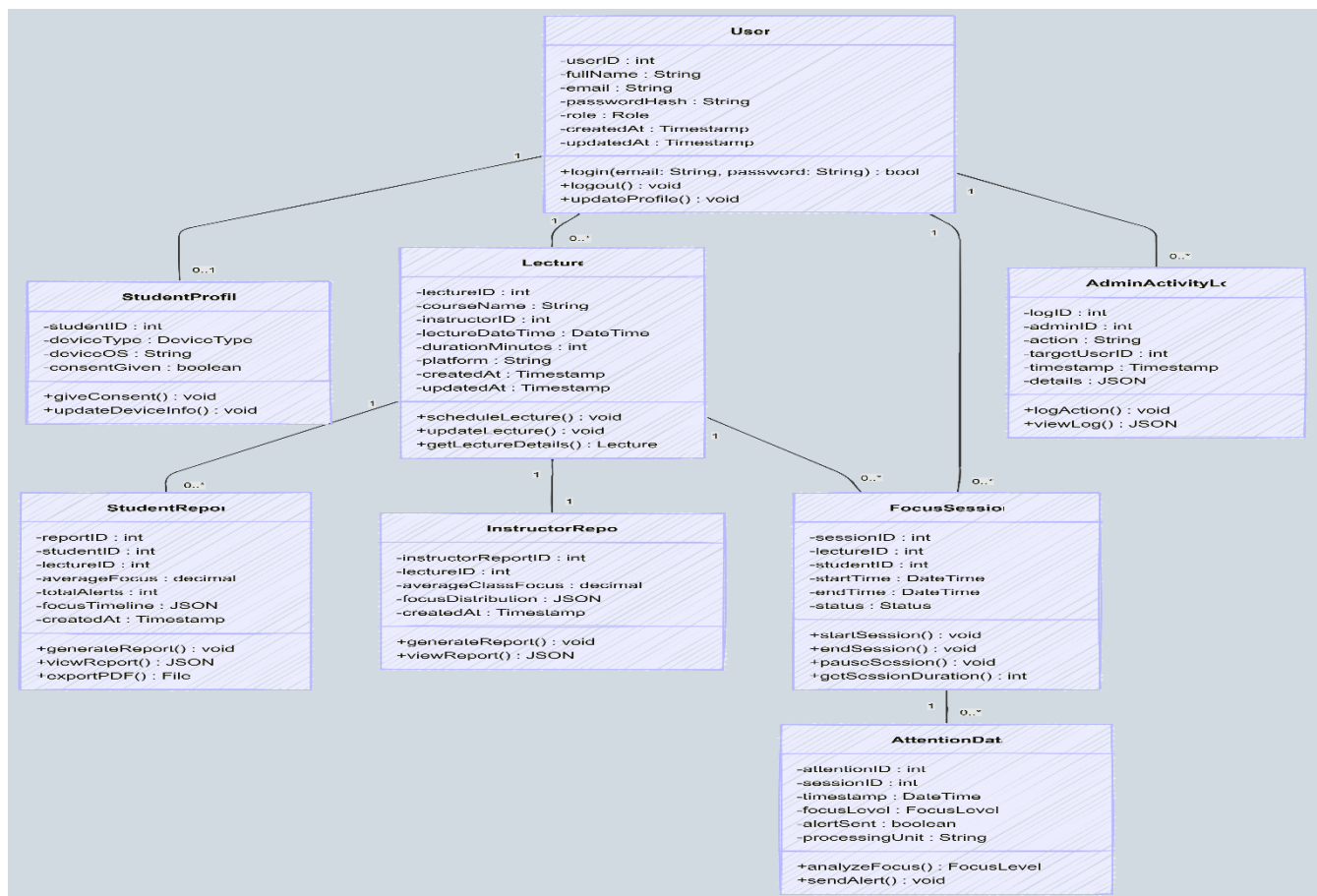
Note: Admin interface is internal; no individual student attention data is displayed.

4. Data Requirements

The EduFocus system track the attention of students during a life lectures and support reporting and feedback for teachers.

4.1 Logical Data Model

The data model represents the main entities, their attributes, and how they relate.



4.2 Reports

The system generates reports for both students and instructors. Reports show attention focus, engagement, session history, and feedbacks without storing raw video or audio.

4.2.1 Student Reports

Purpose: help student to track their attention and focus over time.

Content:

Attention over session	Bar chart showing attention % per session.
Attention distribution	Pie chart showing High-Medium-Low attention.
Summary cards	Average attention (%), total duration, Focus score (0-100).
Personalized feedback	Suggestions and achievements (e.g., “your attention improved by 15%).

Interaction:

- Access via student dashboard.
- Option to download the report.

Constraint:

- No raw video-audio stored.
- Reports generated in real time.

4.2.2 Instructor Reports

Purpose: Provide analytics on student engagement and lecture performance.

Content:

Session information	Course name, total sessions, durations, average students.
Session comparison	Bar charts of attention-engagement per session.
Attention over time	Line chart showing attentions trends.

Engagement summary	Overall attention %, peak engagement, improvement.
Insights & Recommendations	Observations and teaching suggestions.

Interaction:

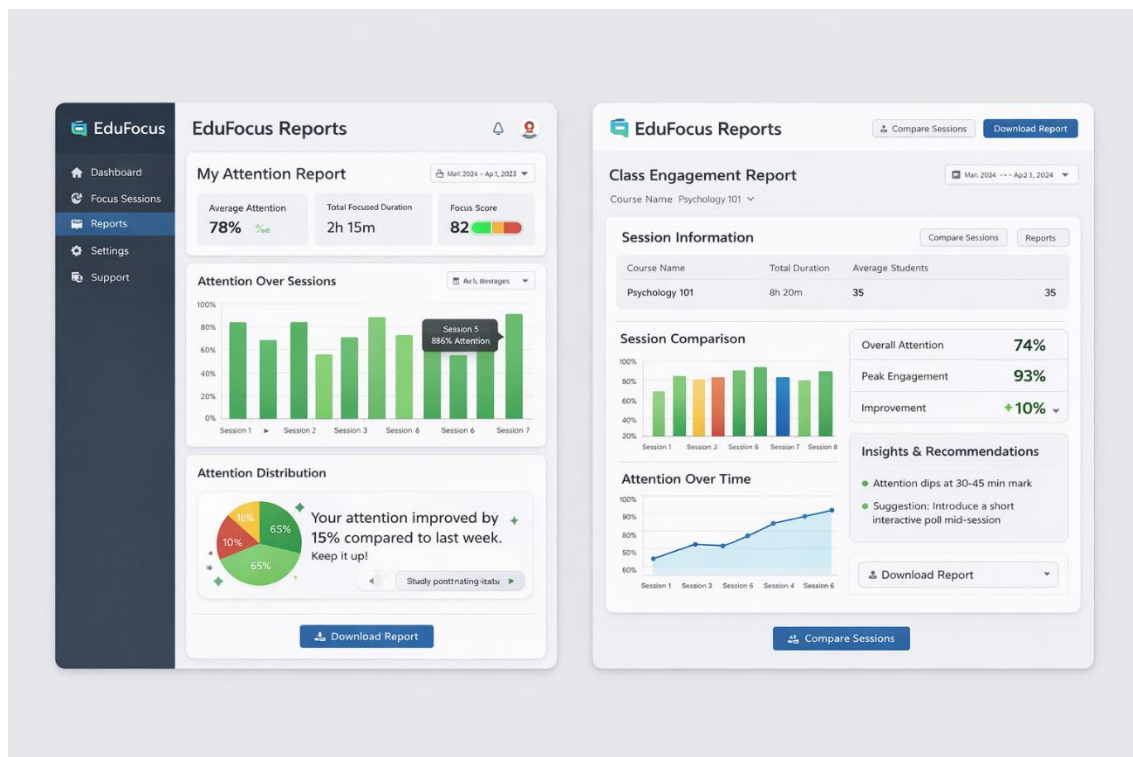
- Access via instructor dashboard.
- Dowland report option.
- Compare sessions, identify attention dips.

Constraints:

- Student data anonymized.
- No raw video-audio.
- Charts auto-scale 0-100%.

Notes:

- Responsive charts for all screen sizes
- Integrates with dashboards
- Ready for future PDF/CSV export



4.3 Data Acquisition, Integrity, Retention, and Disposal

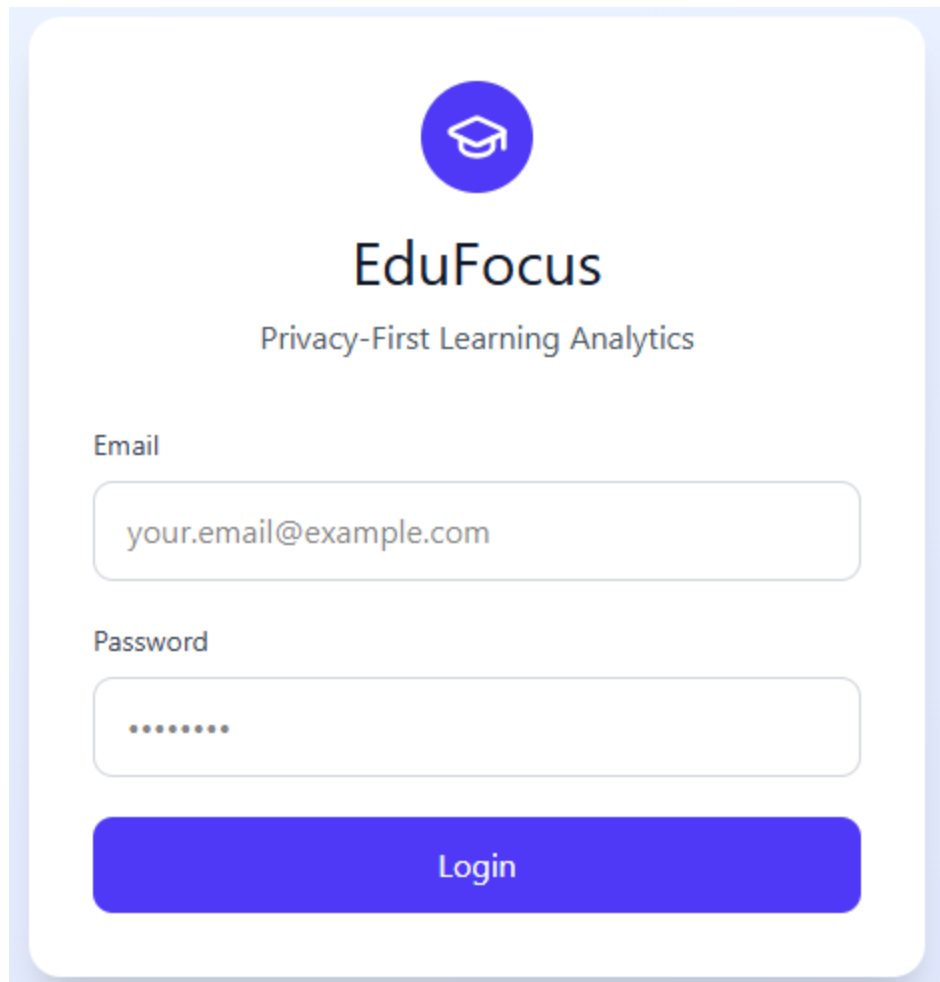
The system collects student interaction and attention data during lectures and basic profile information during registration. Only processed attention data is stored; no raw video or audio is saved. Student data is protected and access is restricted. Attention data is kept for one academic semester, reports are stored for two academic years (Data retention is for academic analysis and reporting compliance), and all temporary data is deleted after report generation. Student data is permanently removed if the account is deleted.

5. External Interface Requirements

As we mentioned, the system provides many special features for both student and instructor.

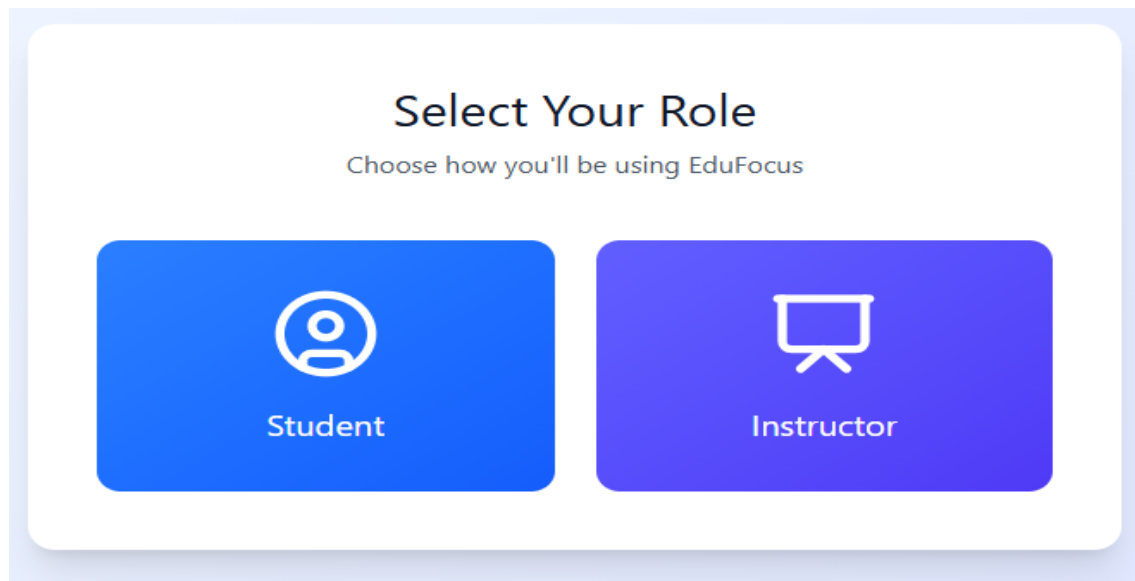
5.1 User Interfaces

Log in: Used by student and instructor to log in to the system in a simple way.



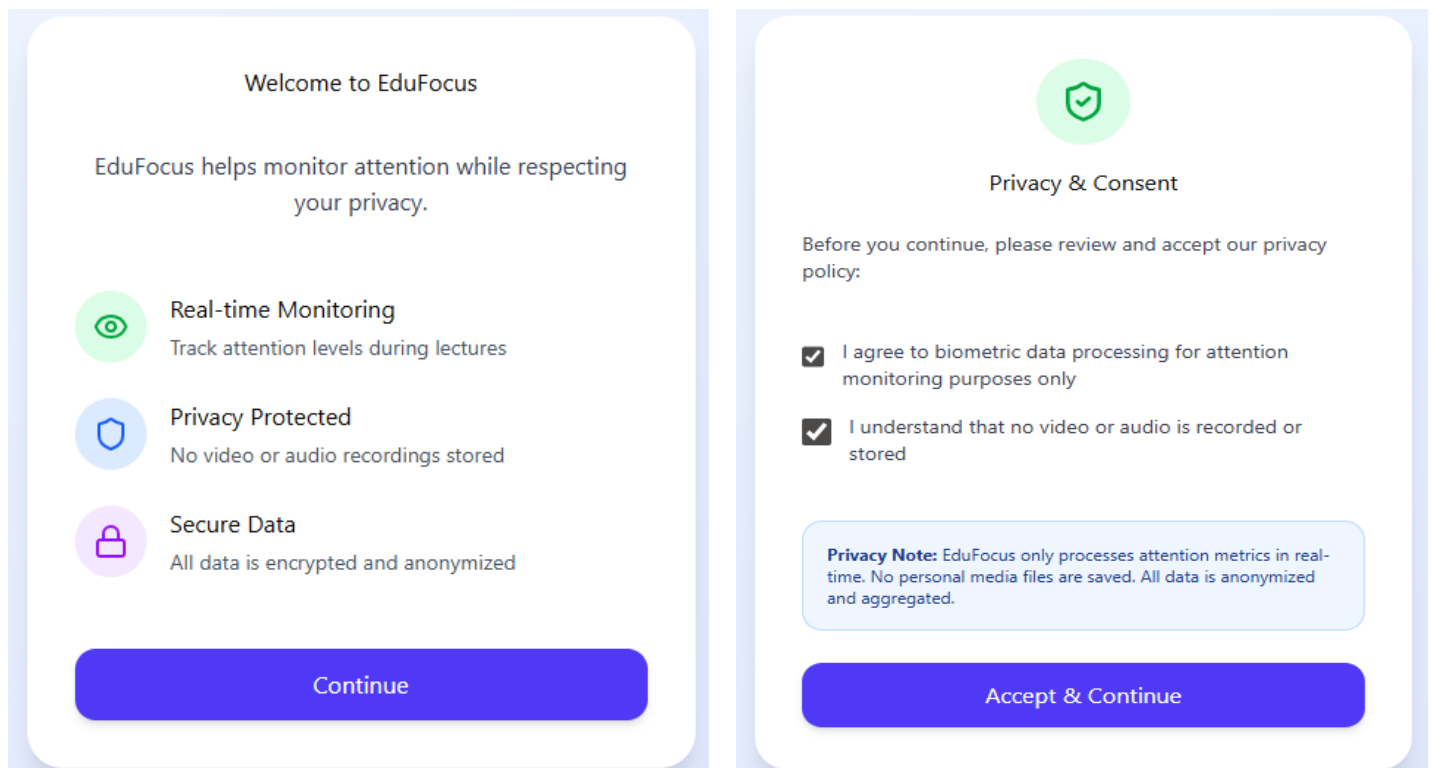
The image shows a login interface for 'EduFocus'. At the top, there is a blue circular icon containing a white graduation cap. Below the icon, the text 'EduFocus' is displayed in a large, bold, black font, followed by 'Privacy-First Learning Analytics' in a smaller, grey font. The interface includes two input fields: 'Email' with a placeholder 'your.email@example.com' and 'Password' with a masked input (dots). A prominent blue 'Login' button is located at the bottom of the form.

I. Select the role: allow the user to choose its role, based on their selection the system display the dashboard suitable to user type.



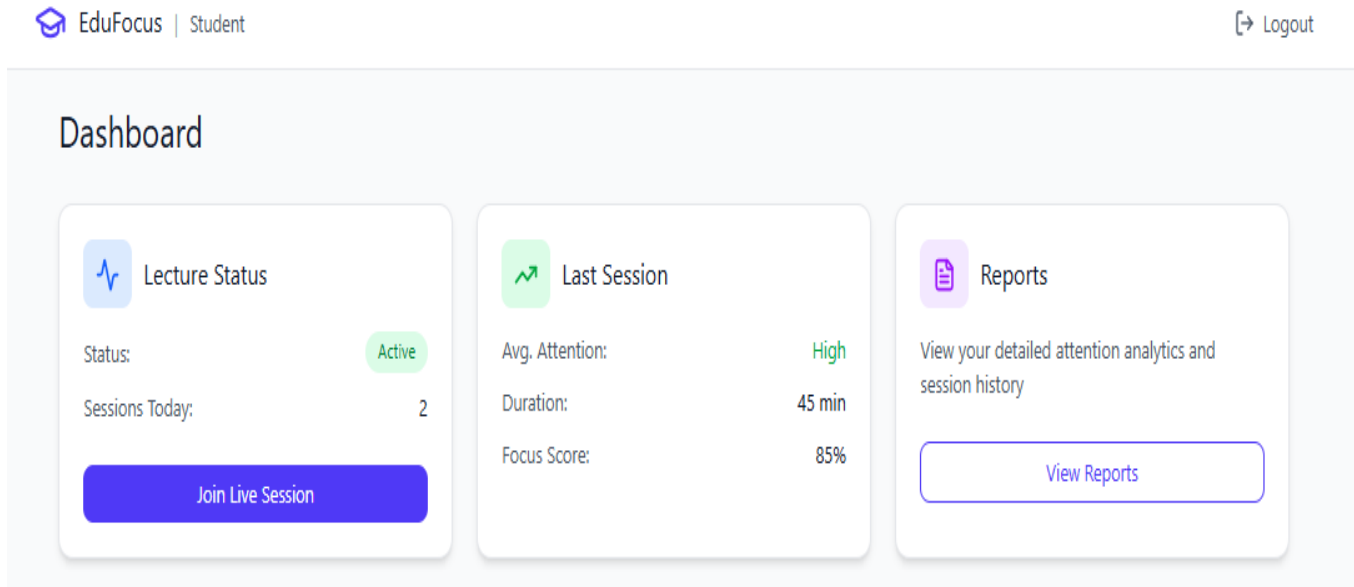
A. Student Role:

1. The system displays the privacy and security information then shows the agreement page.



After agreeing, the student can continue to the main dashboard.

2. Then the system displays the main dashboard to allow the student to start a session and view the summary of their last session.



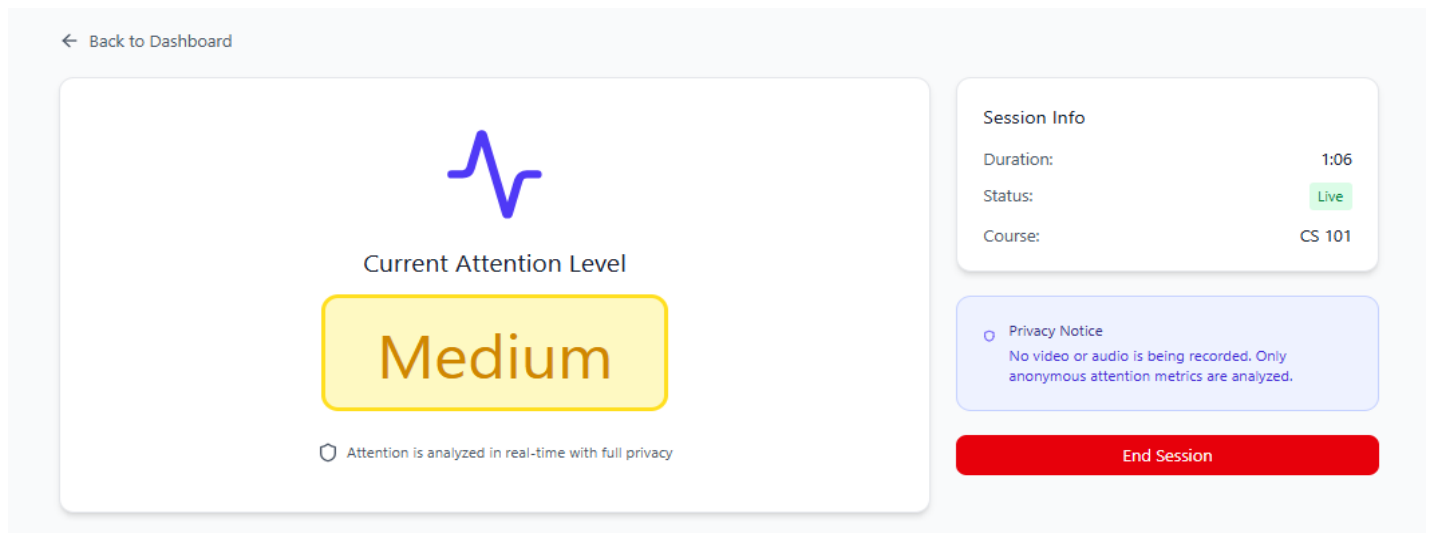
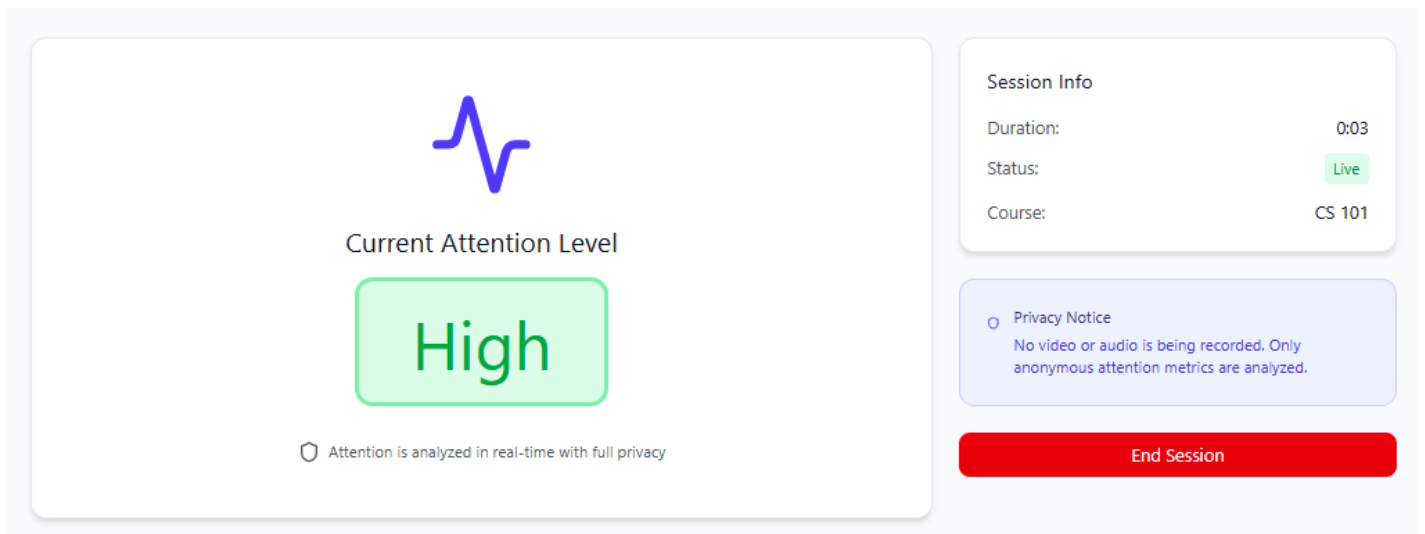
3. When the student joins the session, the system displays a monitoring screen to track the student attention level.

Attention level:

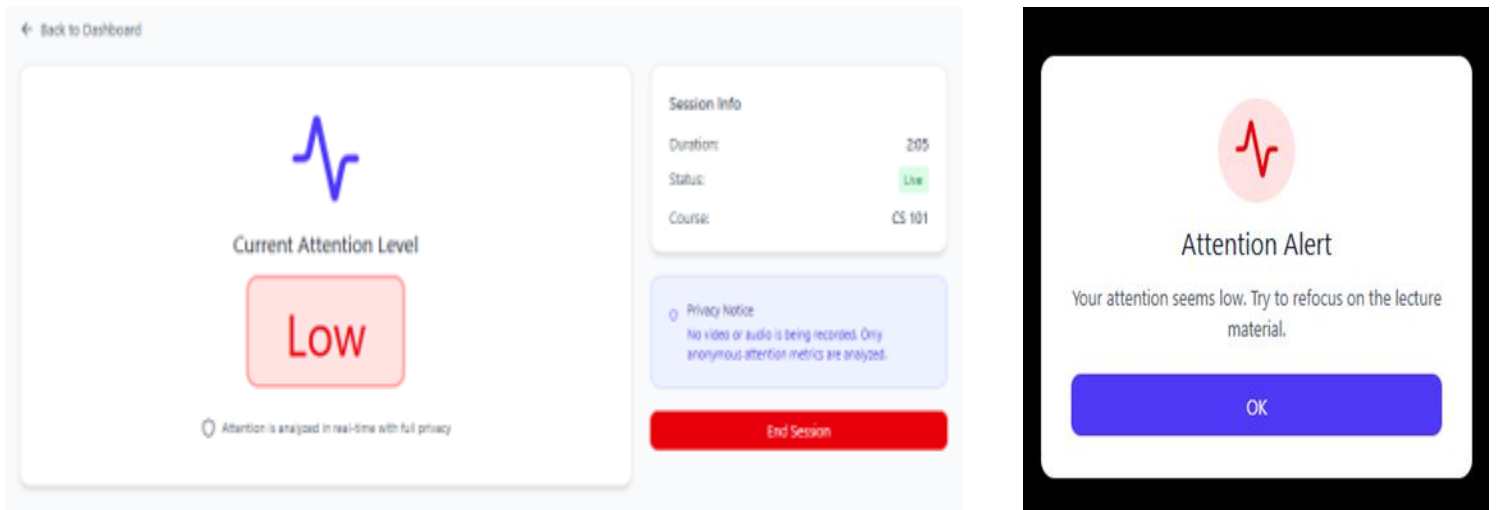
Feature	High	Medium	Low
Eys	Open, looking at the screen	Half open, sometimes looking away	Closed or looking away
Head Position	Straight	A little tilted	Bent down or very tilted
Hand Movement	Small or no movement	Some movement	Distracted
Score	1.0-0.7	0.7-0.4	0.4-0
Attention level	High	Medium	Low

Note:

- Each Feature gets a score from 0-1 depending on what the student doing.
- Overall score = average of all features in one frame. (Frame is a single picture taken from a video)
- Scores are collected over a short time (like 30 second - 1 minute) to find the student's current attention level.



4. When the attention is low a gentle notification is displayed to the student on the screen to remind him to refocus on the lecture.



When the lecture's end, the student can close the session and the system generates a session summary.

5. The student can view their attention report.



B. Instructor Role:

1. The system displays the privacy and security information as student role, then the system display aggreging page

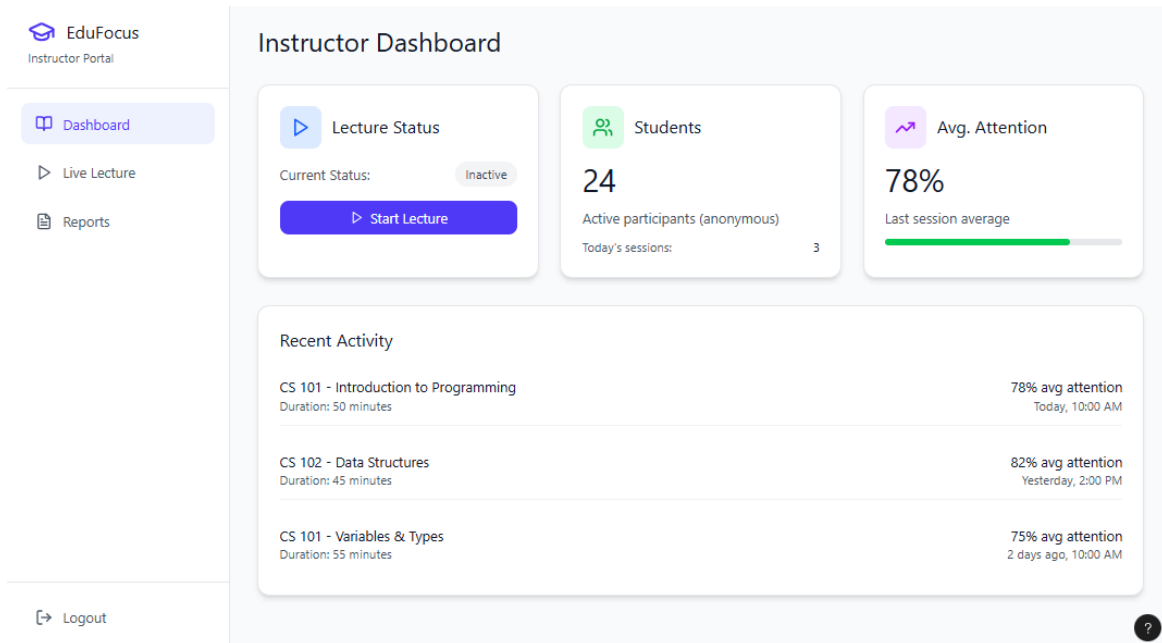
Privacy & Consent
Please review and accept our privacy terms

- ✓ **No Recording:** No video or audio is recorded at any time
- ✓ **Local Processing:** All biometric analysis happens on your device
- ✓ **Anonymous Data:** Only aggregated attention metrics are stored
- ✓ **Full Control:** You can stop analysis at any time

☒ I agree to biometric data processing as described above. I understand that no video or audio will be recorded, and only anonymous attention metrics will be analyzed.

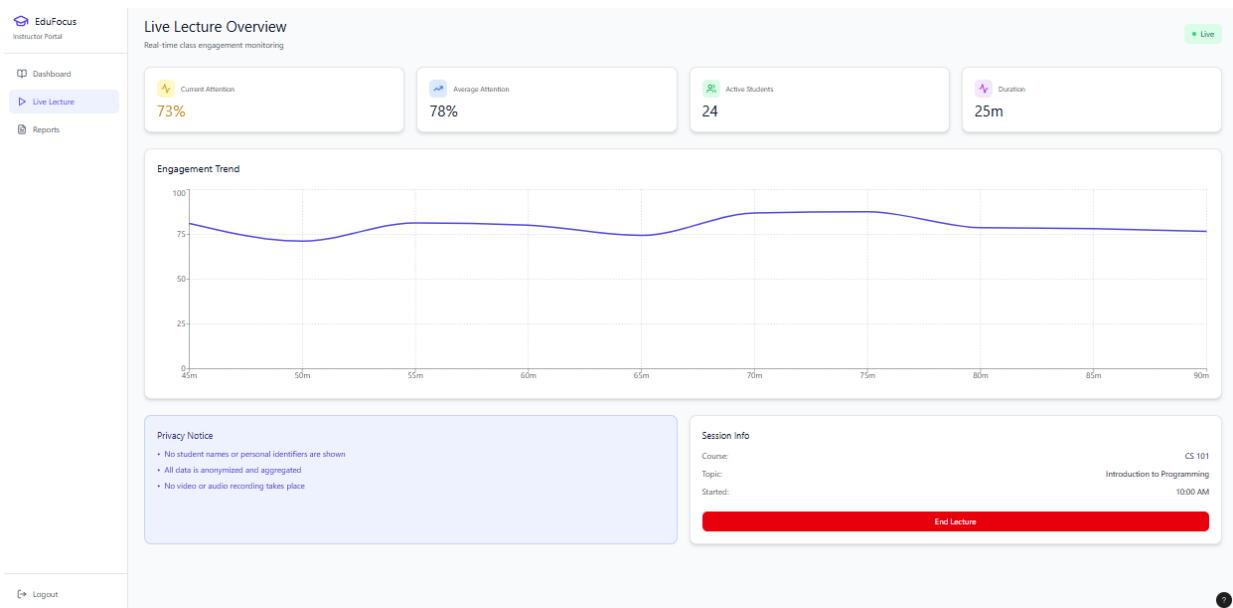
Accept & Continue

- After the instructor agrees to the constraints, the main dashboard is displayed as shown below.



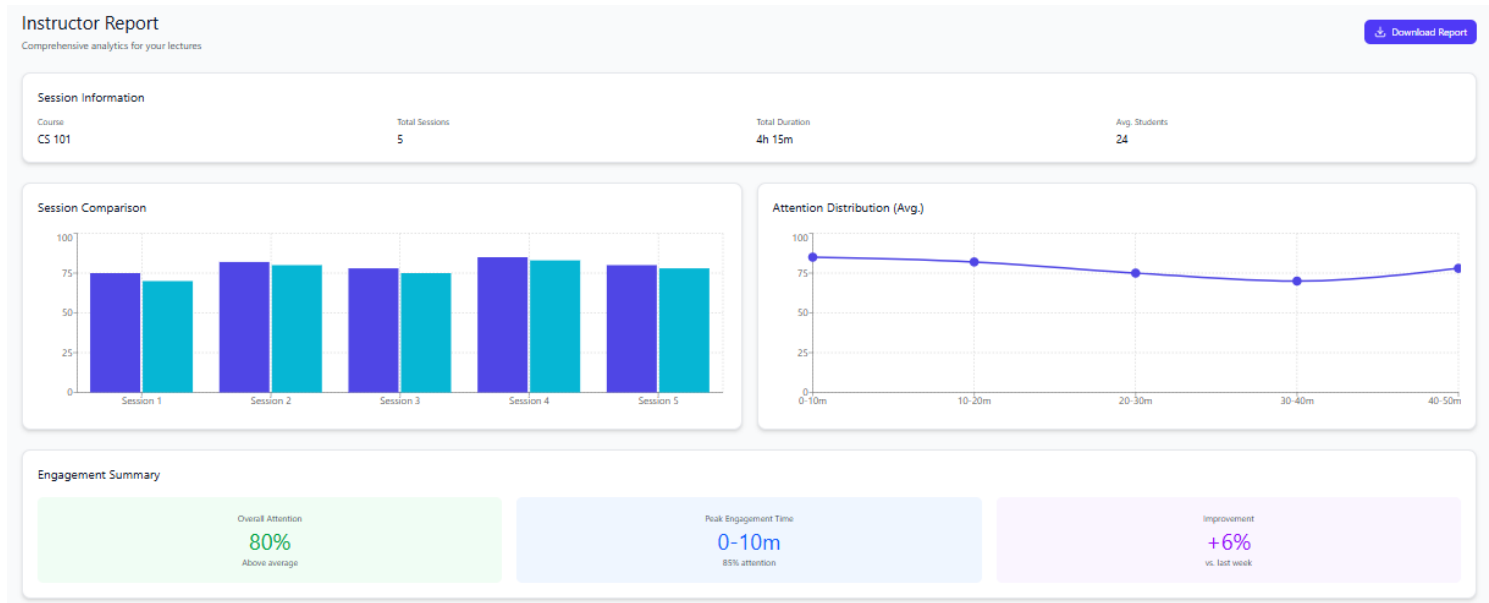
The instructor can start a live lecture and monitor the overall attention level of students from the dashboard.

- When the instructor starts a live lecture, the system displays a live lecture screen with real-time attention statistic.



The instructor can close the live lecture after the lecture is end.

4. The system displays a detailed report that shows session comparisons, attention distribution, and overall engagement analytic.



In summary, Edufocus user interface are designed to be simple, role-based, and privacy, supportin both students and instructors during the lectures.

5.2 Software Interfaces

I. External Services & Data Exchange

External services	Purpose	Input data	Output data	Security protocol
Zoom API	Access live lectures	Lecture request	Lecture ID, Time, Attendance	HTTPS, OAuth
Google Calendar API	Sync lecture schedule	User calendar access	Event ID, Date, Time	HTTPS, OAuth
Notification server	Send alerts	Attention score	Real-time alerts	HTTPS

II. Software Logic & Libraries

Component	Purpose	Processing type	Notes
TensorFlow.js	Attention analysis	Local processing	Video analysis runs on user devices
React.js	User interface	Client-side	Display alerts and dashboard
Node.js	Backend logic	Server-side	Handles APIs and Database communication

III. Data processing (Attention detection)

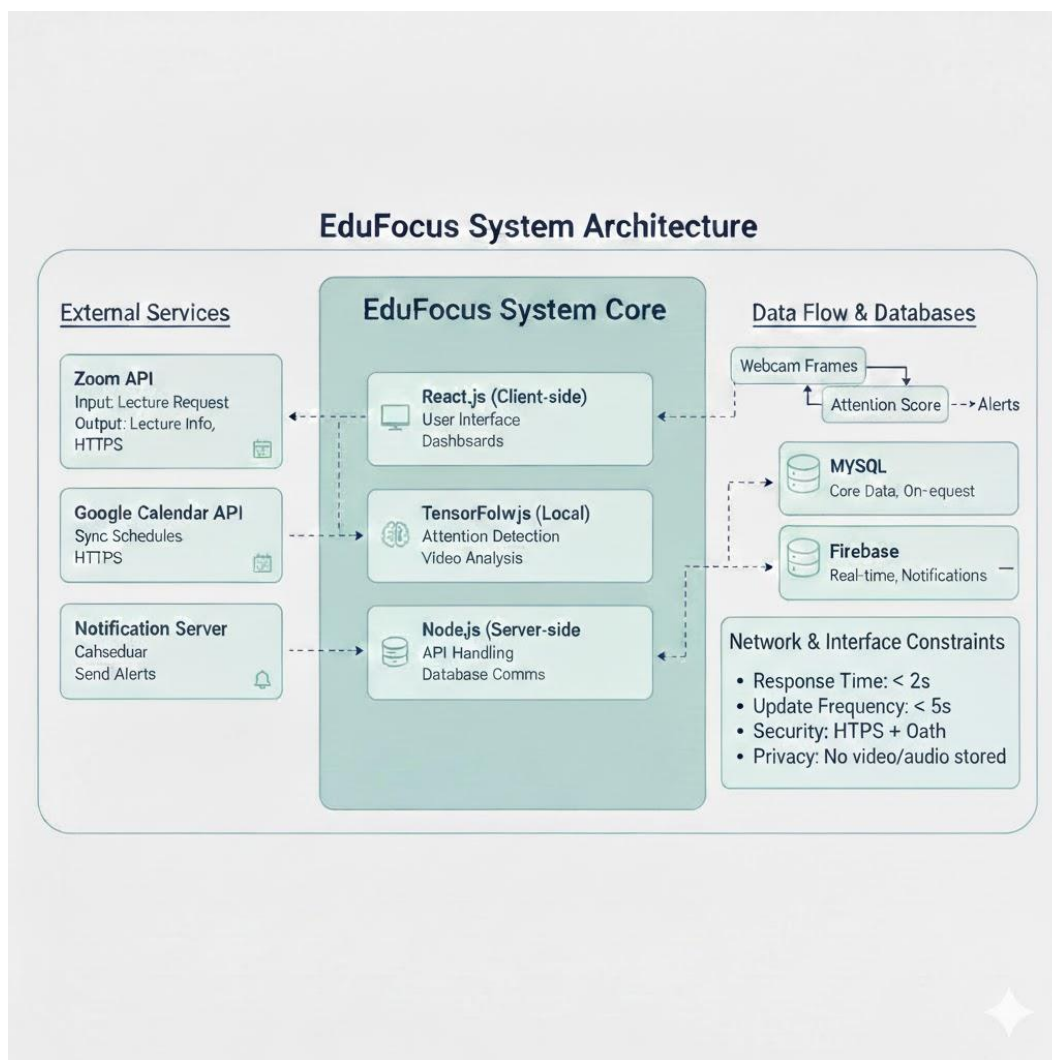
Stage	Data type
Input	Webcam frames
Processing	Eye & gaze analysis using TensorFlow.js
Output	Attention score
Action	Trigger real-time alert if attention is low

IV. Databases

Database	Purpose	Data stores	Sync type
MySQL	Core system data	Student information, scores, attendance	On request
Firebase	Real-time features	Session status, notifications	Real-time sync

V. Network & Interface Constraints

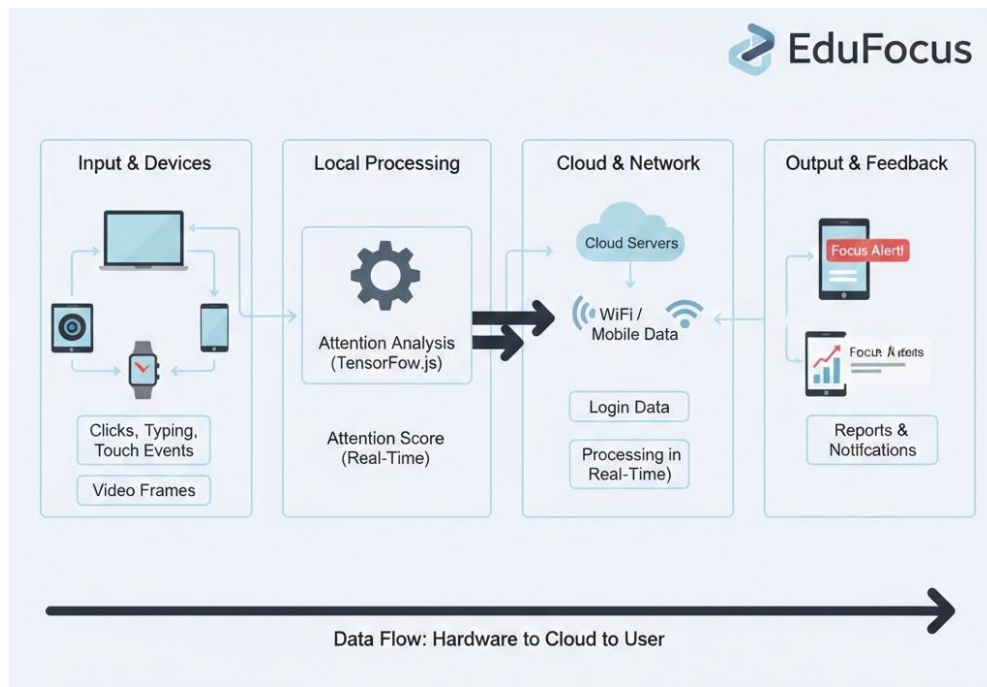
Constraint	Description
Response time	API responses < 2 seconds
Update frequency	Attention analysis every 5 seconds
Security	HTTPS encryption and OAuth authentications
Privacy	No video or audio data stored



5.3 Hardware Interfaces

The system uses common devices and standard hardware.

Interface	Devices	Hardware	Input Data	Output Data	Purpose
User devices	Laptop, Desktop, tablet, mobile, smartwatch.	Camera, Keyboard, Mouse, Touch, Screen, Sensor	Clicks, typing, touch events	Ui feedback, focus alerts	Collect user interaction data
Camera	Laptop or external webcam.	Webcam lens	Video frames	Attention score	Detect student attention.
Network & Server	Cloud servers / internet.	Wi-Fi / mobile data modem.	Attention score, login data	Reports, notifications	Transfer & process data in real time.



6. Quality Attributes

6.1 Usability

EduFocus is made to be easy and simple for all user to use. The main task like logging, join a class and checking the attention are easy to learn. The system is clear, consistent, and work well on laptops, phones, and smartwatches. It helps user to avoid mistakes and shows simple message if the attention is low or distracted.

6.2 Performance

The system aims to deliver a high-performance, seamless experiences by focusing on:

Speed and Reliability	The system is fast and works well, running smoothly on different devices even many users are online.
Efficient process	Tasks are easy and quickly to do, helping users to finish their work without any failures or delays.
User experience	The system has clear and logical screen making it easy to use, even for new users.
Data security	User information is kept safe and private using secure method.
Technical support	The help is available if users have a problem or questions during use the system.

6.3 Security

Security in EduFocus includes protecting user data and privacy, ensuring only the authorized user can access the information.

ID	Rule	Description
SP-1	Data Access Control	Only the authorized user can access to the system.
SP-2	Anonymity	Students' information not visible to the instructor.
SP-3	Data Protection	All users' data is protected from loss, and unauthorized changes.
SP-4	Ethical constraint	No video or audio is recorded or saved.

These measures help user to feel safe and keep their personal information secure.

6.4 Safety

The system focuses on keeping users safe, and making sure the system work correctly during the lectures.

Aspect	Description
Data validation	The system checks all inputs to avoid any error or wrong data.
Privacy	Users' information is kept private and don't share without permissions.
Authorized Access	Only registered user can access the system and use its features.
Data Security	All data is kept safe with secure methods and backups.
Notifications	The system shows alerts and messages to help user avoid any mistake.

6.5 Accuracy

EduFocus tracks student attention carefully to make sure the data is correct and reliable.

Data accuracy	The system measures the student attentions level correctly.
Real-Time Tracking	Attention is tracked live during the lecture,
Error Handling	The system fixed the error and mistakes that happen or tells the user if something goes wrong.
Reports	Reports show what the student behavior during the session

Accuracy is the most important because the system depends on it.

6.6 Additional Quality Attributes

Quality Attribute	Description
Availability	The system works during lecture.
Reliability	The system is stable and can be trusted during the use.
Compatibility	Works on laptops, phones, smartwatches, and different platforms.

Appendix A: Glossary

Term	Definition
EduFocus	A web system that helps track student attention and improve engagement during online or blended university lectures.
SRS	A document that explains all the system requirements, including what the system should do and its quality expectations.
LMS	A platform used by universities to deliver courses and manage learning content.
API	A set of rules that allows different software to communicate and share data.
Zoom API	An interface to connect with Zoom and access lecture data.
Google Calendar API	An interface that allows the system to access Google Calendar events.
OAuth	A secure method to log in or access data without sharing passwords.
React.js	A tool used to build the system's web interface in a clear and user-friendly way.
Node.js	A server-side environment used to process data and communicate with databases.
TensorFlow.js	A library used to analyze student attention from webcam input directly on the device.

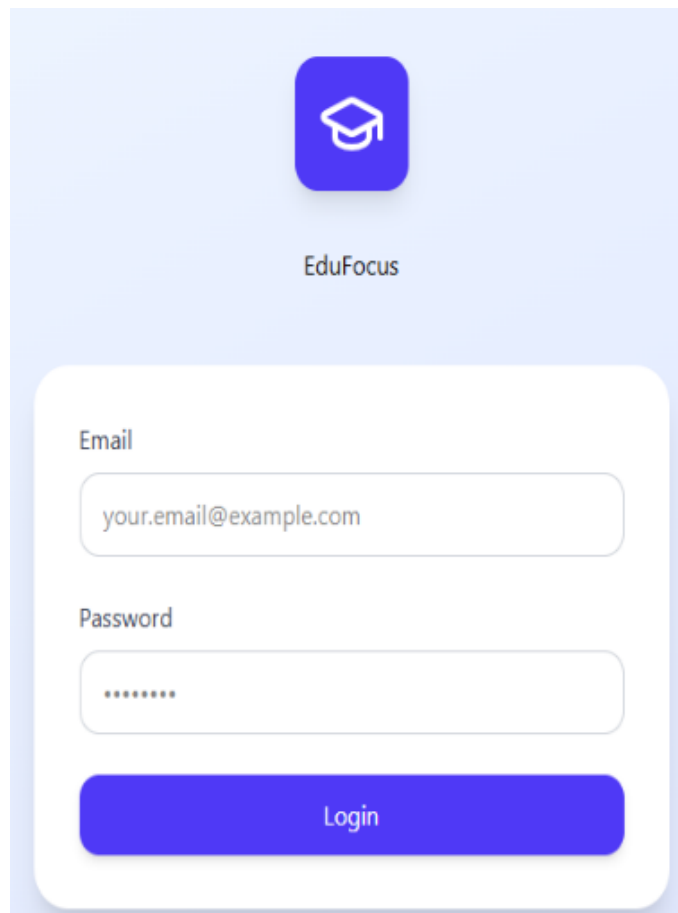
Real-Time Attention Monitoring	Watching a student's focus during a live lecture without saving video or audio.
Attention Score / Focus Index	A number or category (High / Medium / Low) that shows how focused the student is.
Alerts / Notifications	Gentle messages to the student when attention drops, helping them refocus.
Student Reports	Reports generated after lectures showing attention trends and performance.
Instructor Dashboard	A view for instructors to monitor overall class attention without identifying individual students.
Aggregated / Anonymized Data	Data combined from multiple students to give general insights while protecting privacy.
Session	A single lecture period during which attention is monitored.
Student Dashboard	The interface where students see their attention level, alerts, and past reports.
Data Export (PDF/CSV)	Ability to download reports for offline review.
Wearable Devices	Devices like smartwatches used to track heart rate or movement.
Webcam / Camera	A camera used to analyze focus in real time without saving images.
Server / Cloud Hosting	Computers that run the system and handle data securely.
Privacy Policy	Rules that protect student data and prevent storing audio or video.
Data Retention / Disposal	How long data is kept and when it is deleted.
Administrator / System Administrator	A user responsible for managing the system, accounts, and settings while ensuring privacy.
Access Control	Restricting access to system functions and data based on user roles.
High / Medium / Low Attention	Classifying student focus based on eye, head, hand movement, and heart rate.
Frame	A temporary image from the webcam used to calculate attention, not stored.

Local Processing	All analysis happens on the student's device, no video is sent to the server.
Session Management	Creating, monitoring, and ending lecture sessions while safely recording attention data.
Alerts Threshold	The attention level below which the system sends a notification.
Aggregated Reports	Reports summarizing attention from multiple students without showing identities.

Appendix B: Analysis Models

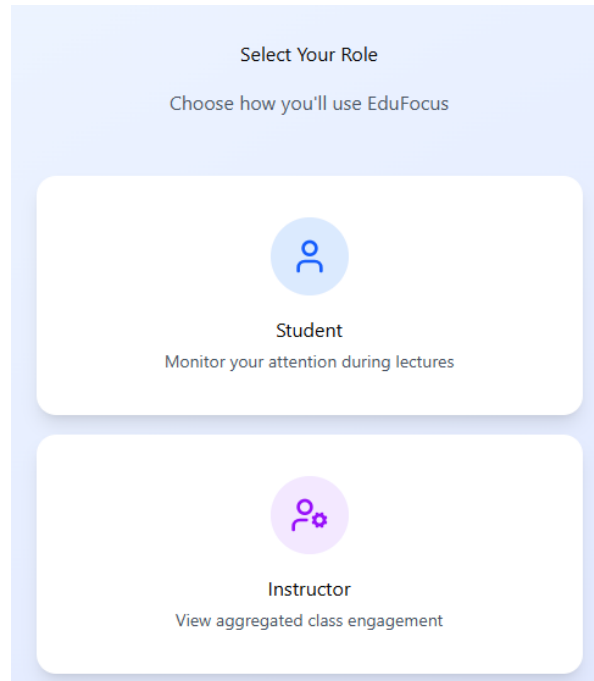
1. Mobile application prototype

- a. **Login screen:** the main interface for logging in to the EduFocus mobile application.



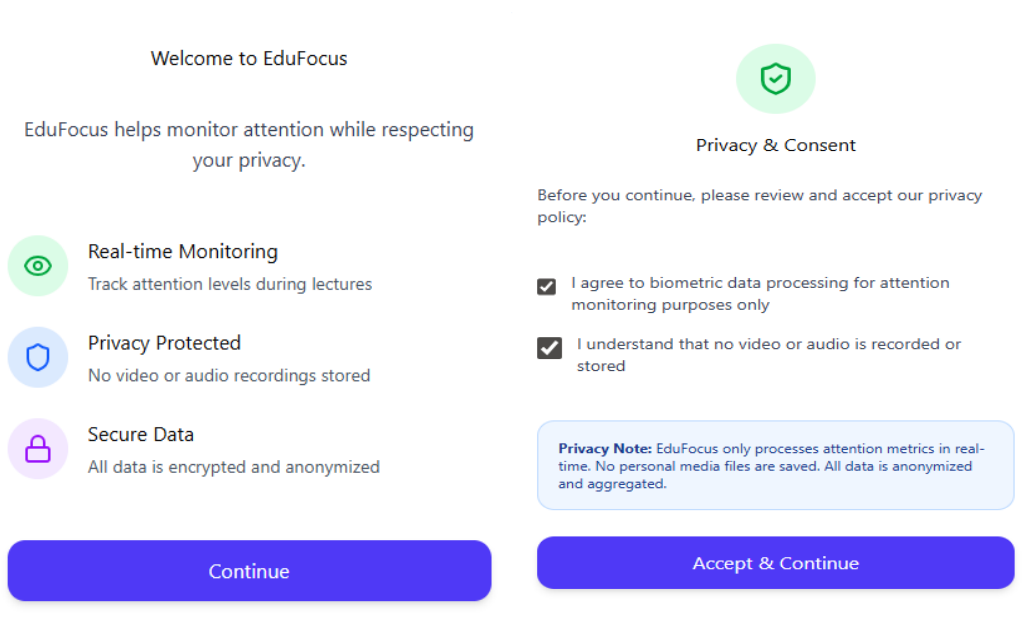
The image shows a mobile application login screen for 'EduFocus'. At the top, there is a blue square icon with a white graduation cap. Below the icon, the text 'EduFocus' is displayed. The main content area is a white rounded rectangle with a light blue background. It contains two input fields: 'Email' with the placeholder text 'your.email@example.com' and 'Password' with a masked password '*****'. Below these fields is a blue rounded rectangle button with the text 'Login'.

- b. Role selection: this screen allows the users to select their role (student or instructor).



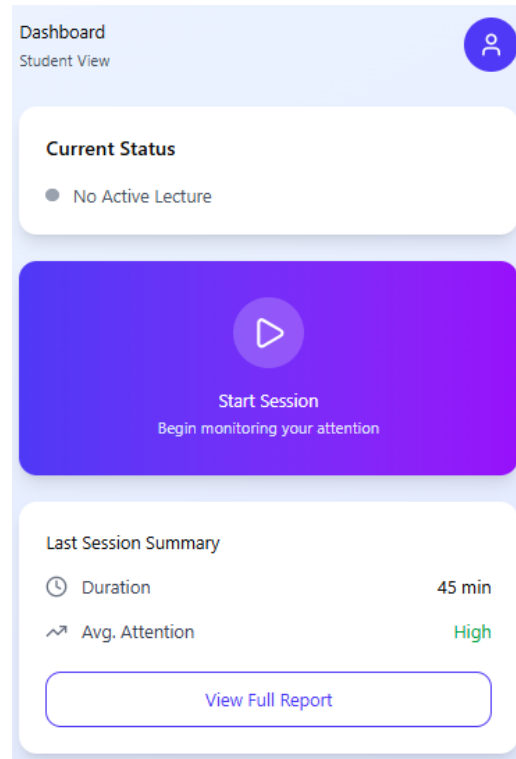
○ **Student role:**

- I. The system displays the privacy and security information then shows the agreement page.

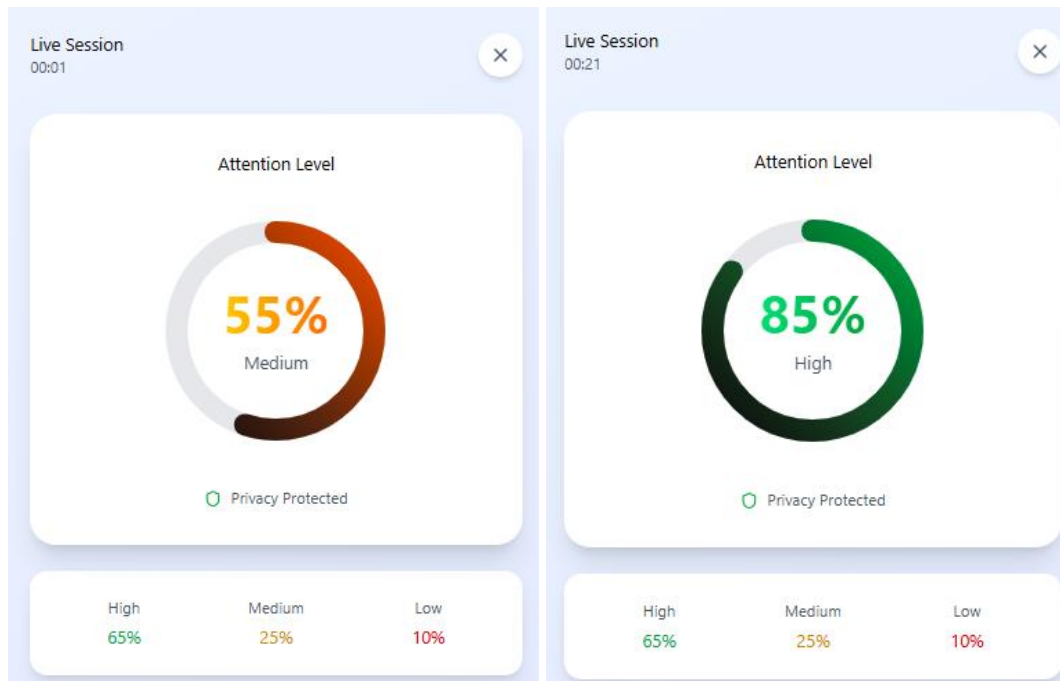


After agreeing, the student can continue to the main dashboard.

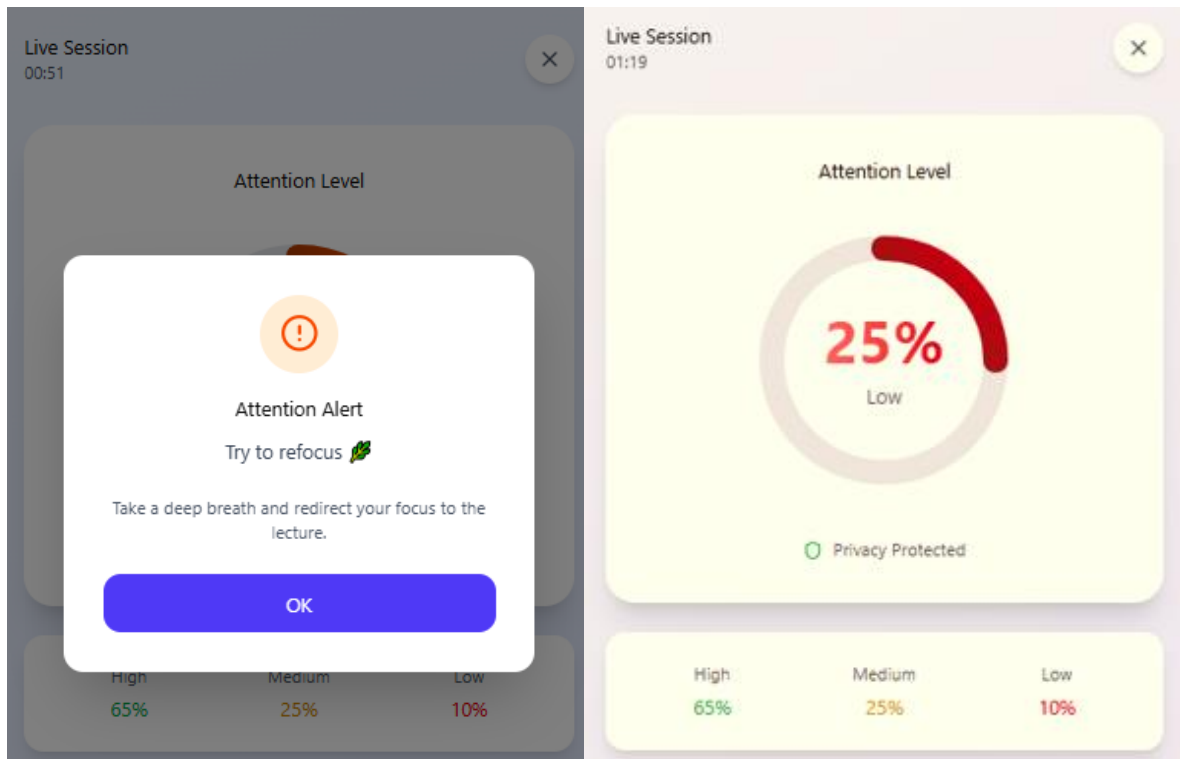
- II. Then the system displays the main dashboard to allow the student to start a session and view the summary of their last session.



- III. When the student joins the session, the system displays a monitoring screen to track the student attention level.

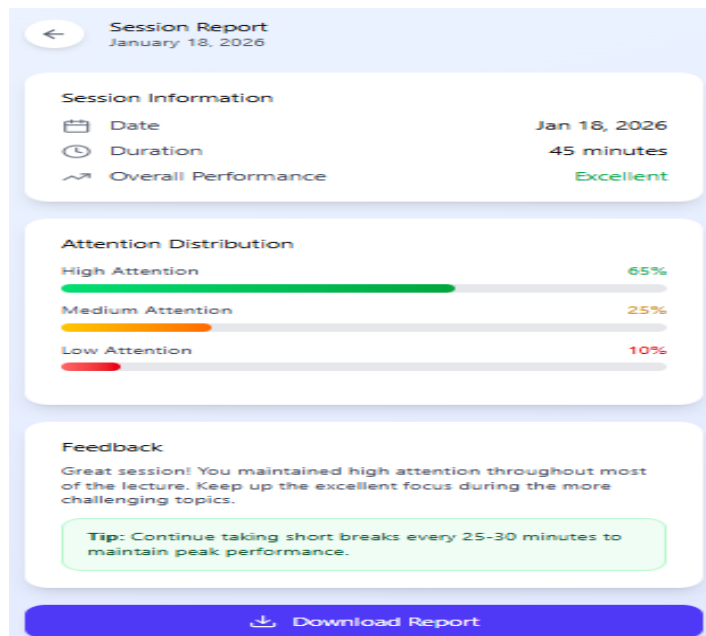


- IV. When the attention is low a gentle notification is displayed to the student on the screen to remind him to refocus on the lecture.

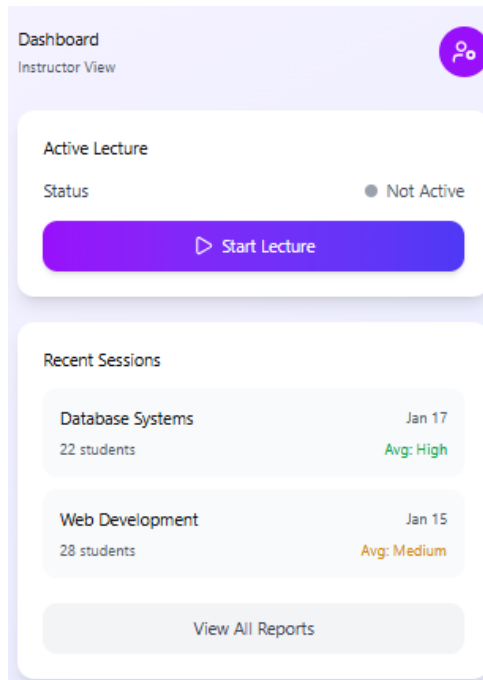


When the lecture's end, the student can close the session and the system generates a session summary.

- V. The student can view their attention report.

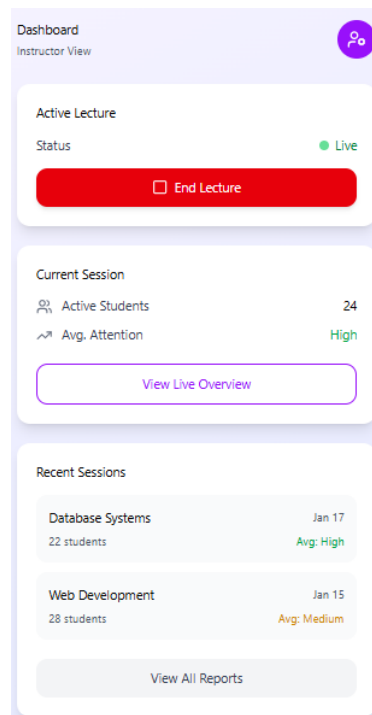


- Instructor role
- 1. The system displays the privacy and security information as student role, then the system display aggreging page.
- 2. After the instructor agrees to the constraints, the main dashboard is displayed as shown below.

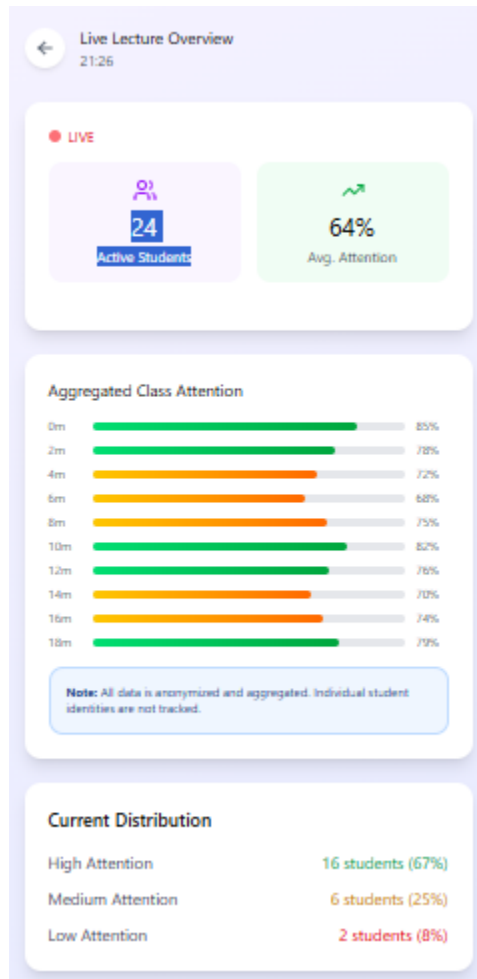


The instructor can start a live lecture and monitor the overall attention level of students from the dashboard.

- 3. When the instructor starts a live lecture, the system displays a live lecture screen with real-time attention statistic.

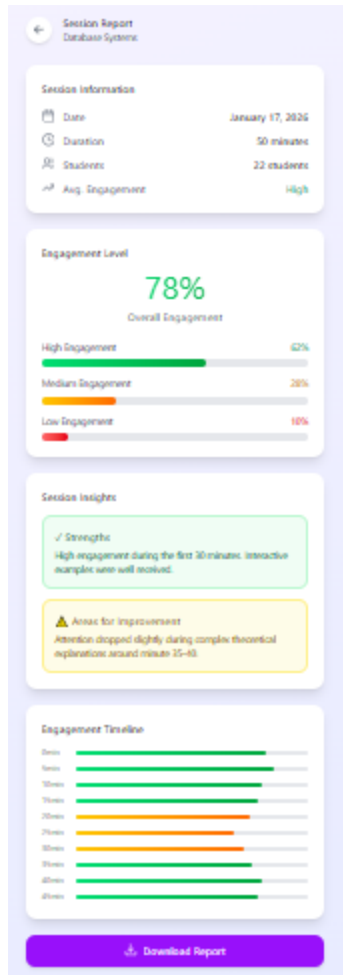


4. The instructor can view live class attention and the number of active students.



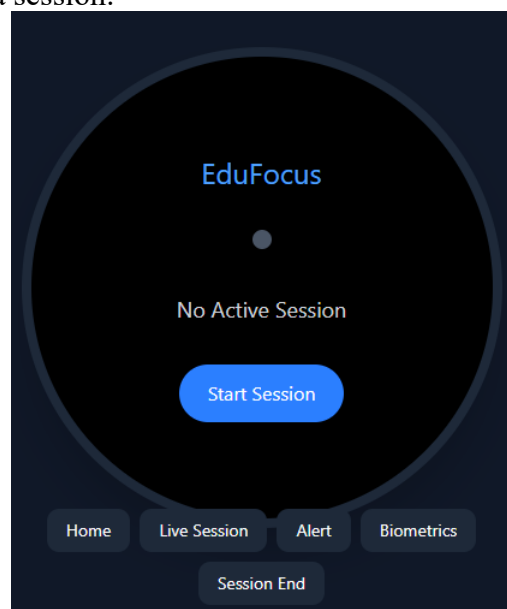
The instructor can close the live lecture after the lecture is end.

5. The system displays a detailed report that shows session comparisons, attention distribution, and overall engagement analytic.

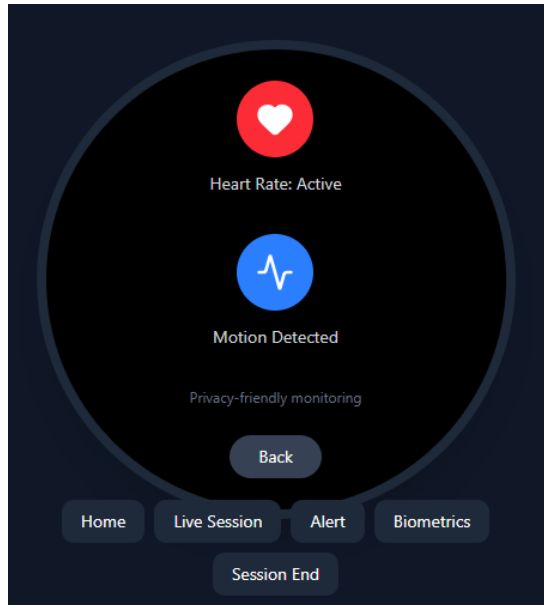


2. Smartwatch Prototype

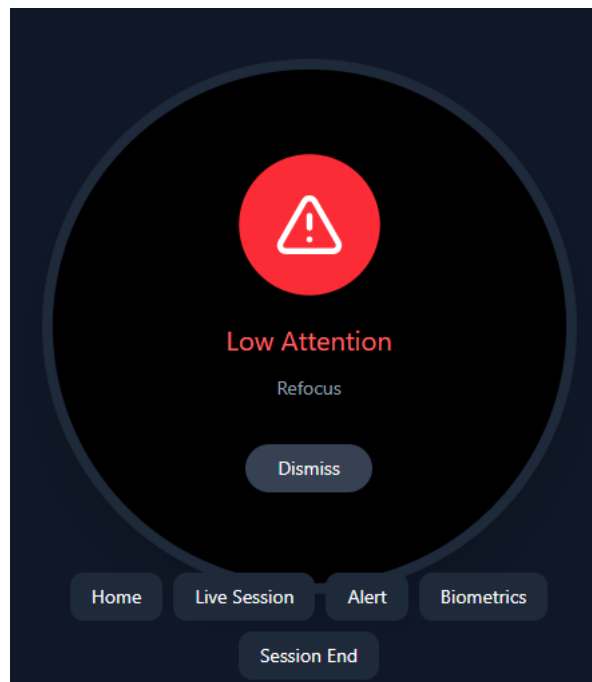
a) The student can start a session.



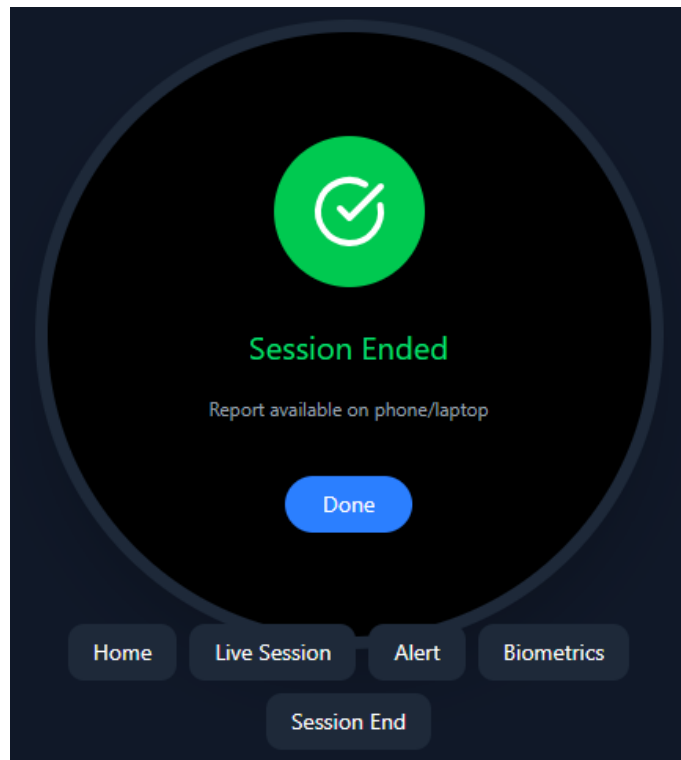
b) After the student start a session, the smartwatch shows heart rate and motion status.



c) After the student's attention drops, the smartwatch shows a low-attention alert.



- d) After the student ends the session, the smartwatch displays a confirmation message indicating that the session has ended and the report is available on the phone or laptop.



3. Attention Scoring Scale

The EduFocus system calculates a Focus Score (Attention Index) to measure how attentive a student is during a lecture. This score combines multiple features collected in real time from the student's device, such as:

- Eyes: Open, looking at the screen, or closed-looking away
- Head Position: Straight, slightly tilted, or bent/very tilted
- Hand Movement: Minimal, moderate, or distracting
- Heart Rate - Motion (if wearable device is used): Normal range, slightly irregular, or signs of restlessness.

Each feature is converted into a score from 0 to 1, where 1 indicates full attention and 0 indicates complete distraction. The overall Focus Score is calculated as the average of all feature scores in a short observation period (called a frame, typically 30–60 seconds).

Attention Level Classification:

- High (0.7 – 1.0): Student is focused and attentive.
- Medium (0.4 – 0.7): Student is partially distracted; occasionally looking away.
- Low (0 – 0.4): Student is distracted, disengaged, or not looking at the lecture.

Notes:

- Scores are updated in real time during the lecture.
- Processing occurs locally on the student's device; no video or audio is stored.
- When the Focus Score falls below a certain threshold, the system sends gentle alerts to help the student refocus.

EduFocus Attention Scoring Scale



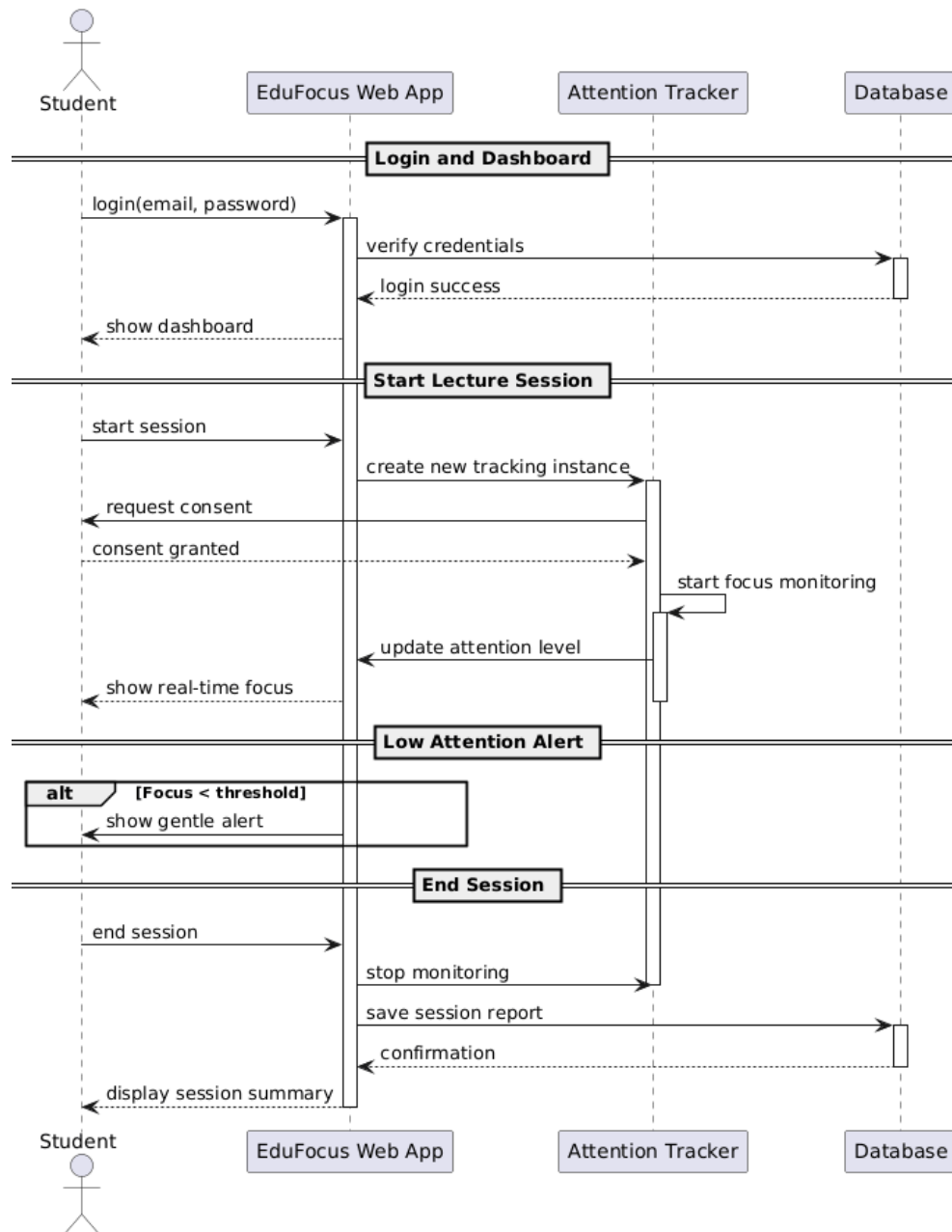
Focus Score = average of Eyes, Head Position, Hand Movement, Heart Rate
Updated in real time, no video/audio stored

The attention scale adopted in EduFocus (High, Medium, Low) is based on prior research that estimates student attention using observable behavioral and visual indicators such as eye gaze direction, head position, and movement patterns, rather than direct cognitive measurement.

This approach is supported by recent studies on attention level estimation in interactive systems (Al-Hmouz et al., 2023).

Appendix C: UML Diagrams and Survey Result

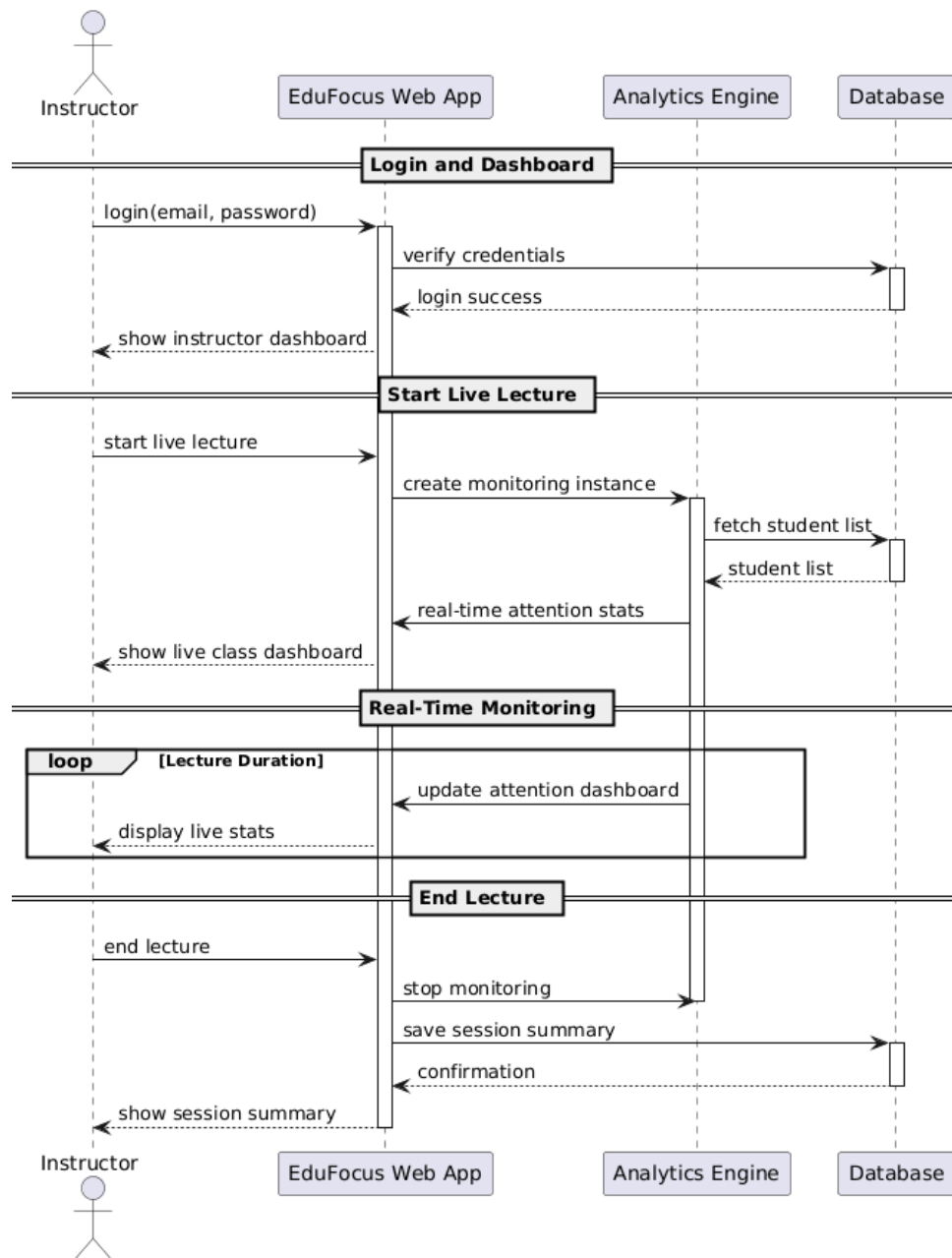
1. Sequence diagram for student



Shows how the student logs in, monitors attention in real time, and gets a session report.

Keeps privacy by not storing any video or audio.

2. Sequence diagram for instructor



Shows how the instructor starts a lecture, monitors class attention, and gets a summary report. Displays only aggregated data, without revealing individual students.

A. Student survey result

Topic	Key Points
Who answered	Undergraduate, Master's, and PhD students
Focus in class	Some students stay focused, some lose focus often
Main problems	Phones, feeling tired, teacher's style, noisy environment
How they cope	Taking notes, asking questions, avoiding phones
Monitoring system	Most students like real-time alerts and reports
Devices used	Phone, smartwatch, laptop, tablet
Notifications	Many want alerts on one or more devices

B. Instructor survey result

Teaching Experience	Learning Mode	Attention Frequency	Common Distractions	Most Used Features
Less than 5 years	Blended learning	Often/Always	Lack of participation, Camera off, Late responses, Poor quiz results, Phone use	Attention reports, Live alerts, Participation analytics, Attendance tracking
5–10 years	Blended/Face-to-face	Often	Lack of participation, Late responses, Poor quiz results, Phone use	Attention reports, Live alerts, Attendance tracking
More than 10 years	Face-to-face / Blended	Often/Always	Lack of participation, Camera off, Late responses, Poor quiz results, Phone use, Side conversations	Attention reports, Live alerts, Participation analytics, Attendance tracking